Chapter 7



(© Tony Freeman/Science Source)

When external loads are placed upon these beams and columns, the loads within them must be determined if they are to be properly designed. In this chapter we will study how to determine these internal loadings.

Internal Forces

CHAPTER OBJECTIVES

- To use the method of sections to determine the internal loadings in a member at a specific point.
- To show how to obtain the internal shear and moment throughout a member and express the result graphically in the form of shear and moment diagrams.
- To analyze the forces and the shape of cables supporting various types of loadings.

7.1 Internal Loadings Developed in Structural Members

To design a structural or mechanical member it is necessary to know the loading acting within the member in order to be sure the material can resist this loading. Internal loadings can be determined by using the ***method of sections***. To illustrate this method, consider the cantilever beam in Fig. 7–1*a*. If the internal loadings acting on the cross section at point *B* are to be determined, we must pass an imaginary section *a–a* perpendicular to the axis of the beam through point *B* and then separate the beam into two segments. The internal loadings acting at *B* will then be exposed and become *external* on the free-body diagram of each segment, Fig. 7–1*b*.

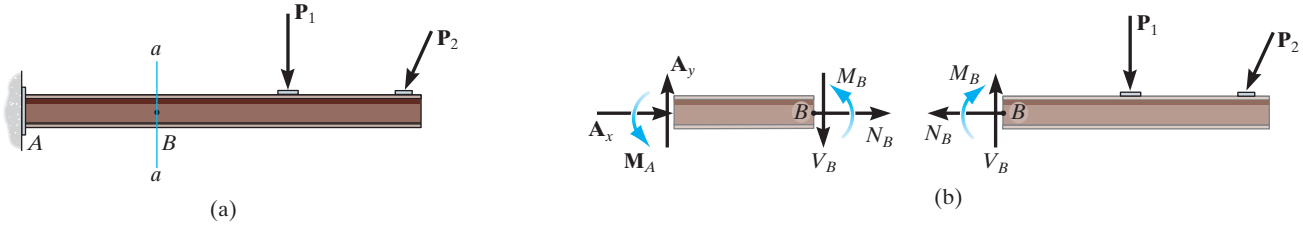
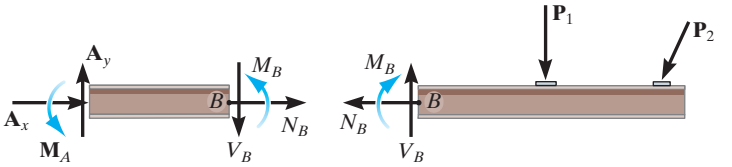


Fig. 7–1





In each case, the link on the backhoe is a two-force member. In the top photo it is subjected to both bending and an axial load at its center. It is more efficient to make the member straight, as in the bottom photo; then only an axial force acts within the member. (© Russell C. Hibbeler)

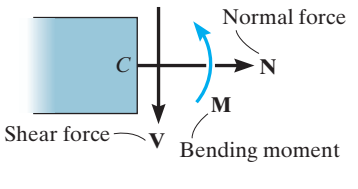


(b)

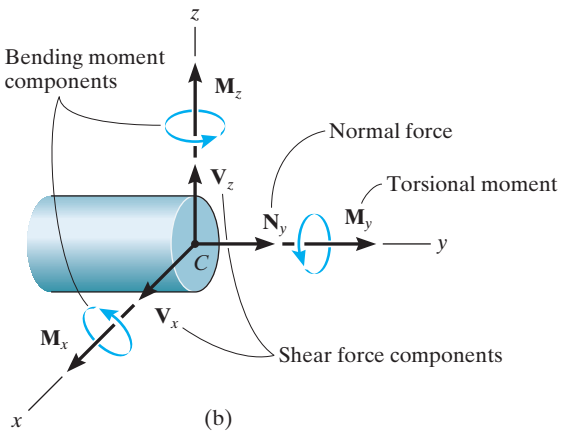
Fig. 7–1 (Repeated)

The force component $\mathbf{N}_B$ that acts *perpendicular* to the cross section is termed the ***normal force***. The force component $\mathbf{V}_B$ that is tangent to the cross section is called the ***shear force***, and the couple moment $\mathbf{M}_B$ is referred to as the ***bending moment***. The force components prevent the relative translation between the two segments, and the couple moment prevents the relative rotation. According to Newton's third law, these loadings must act in opposite directions on each segment, as shown in Fig. 7–1*b*. They can be determined by applying the equations of equilibrium to the free-body diagram of either segment. In this case, however, the right segment is the better choice since it does not involve the unknown support reactions at A. A direct solution for $\mathbf{N}_B$ is obtained by applying $\Sigma F_x = 0$, $\mathbf{V}_B$ is obtained from $\Sigma F_y = 0$, and $\mathbf{M}_B$ can be obtained by applying $\Sigma M_B = 0$, since the moments of $\mathbf{N}_B$ and $\mathbf{V}_B$ about B are zero.

In two dimensions, we have shown that three internal loading resultants exist, Fig. 7–2*a*; however in three dimensions, a general resultant internal force and couple moment resultant will act at the section. The x, y, z components of these loadings are shown in Fig. 7–2*b*. Here $\mathbf{N}_y$ is the *normal force*, and $\mathbf{V}_x$ and $\mathbf{V}_z$ are *shear force components*. $\mathbf{M}_y$ is a *torsional or twisting moment*, and $\mathbf{M}_x$ and $\mathbf{M}_z$ are *bending moment components*. For most applications, these *resultant loadings* will act at the geometric center or centroid (C) of the section's cross-sectional area. Although the magnitude for each loading generally will be different at various points along the axis of the member, the method of sections can always be used to determine their values.



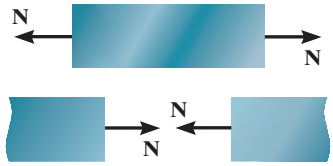
(a)



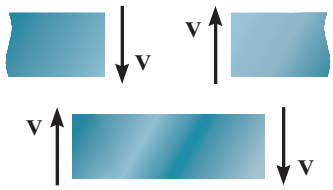
(b)

Fig. 7–2

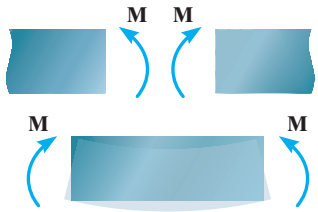
Sign Convention. For problems in two dimensions engineers generally use a sign convention to report the three internal loadings **N, V,** and **M**. Although this sign convention can be arbitrarily assigned, the one that is widely accepted will be used here, Fig. 7–3. The normal force is said to be positive if it creates *tension*, a positive shear force will cause the beam segment on which it acts to rotate clockwise, and a positive bending moment will tend to bend the segment on which it acts in a concave upward manner. Loadings that are opposite to these are considered negative.



Positive normal force



Positive shear



Positive moment

Fig. 7–3

Important Point

- There can be four types of resultant internal loads in a member. They are the normal and shear forces and the bending and torsional moments. These loadings generally vary from point to point. They can be determined using the method of sections.

Procedure for Analysis

The method of sections can be used to determine the internal loadings on the cross section of a member using the following procedure.

Support Reactions.

- Before the member is sectioned, it may first be necessary to determine its support reactions.

Free-Body Diagram.

- It is important to *keep* all distributed loadings, couple moments, and forces acting on the member in their *exact locations*, *then* pass an imaginary section through the member, perpendicular to its axis at the point where the internal loadings are to be determined.
- After the section is made, draw a free-body diagram of the segment that has the least number of loads on it, and indicate the components of the internal force and couple moment resultants at the cross section acting in their positive directions in accordance with the established sign convention.

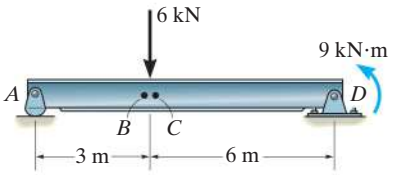
Equations of Equilibrium.

- Moments should be summed at the section. This way the normal and shear forces at the section are eliminated, and we can obtain a direct solution for the moment.
- If the solution of the equilibrium equations yields a negative scalar, the sense of the quantity is opposite to that shown on the free-body diagram.



The designer of this shop crane realized the need for additional reinforcement around the joint at *A* in order to prevent severe internal bending of the joint when a large load is suspended from the chain hoist. (© Russell C. Hibbeler)

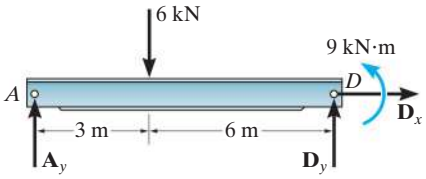
Determine the normal force, shear force, and bending moment acting just to the left, point B, and just to the right, point C, of the 6-kN force on the beam in Fig. 7–4*a*.



(a)

SOLUTION

Support Reactions. The free-body diagram of the beam is shown in Fig. 7–4*b*. When determining the *external reactions*, realize that the 9-kN · m couple moment is a free vector and therefore it can be placed *anywhere* on the free-body diagram of the entire beam. Here we will only determine $\mathbf{A}_y$, since the left segments will be used for the analysis.

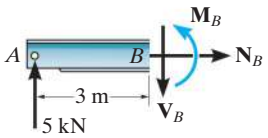


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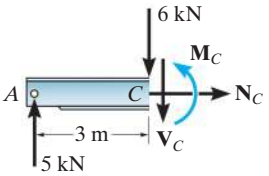
$$\circlearrowright +\Sigma M_D = 0; \quad 9\ \text{kN}\cdot\text{m} + (6\ \text{kN})(6\ \text{m}) - A_y(9\ \text{m}) = 0$$

$$A_y = 5\ \text{kN}$$

Free-Body Diagrams. The free-body diagrams of the left segments AB and AC of the beam are shown in Figs. 7–4*c* and 7–4*d*. In this case the 9-kN · m couple moment is *not included* on these diagrams since it must be kept in its *original position* until *after* the section is made and the appropriate segment is isolated.



(c)



(d)

Fig. 7–4

Equations of Equilibrium.

Segment AB

$$\pm\rightarrow \Sigma F_x = 0; \quad N_B = 0 \qquad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \quad 5\ \text{kN} - V_B = 0 \quad V_B = 5\ \text{kN} \qquad \textit{Ans.}$$

$$\circlearrowright +\Sigma M_B = 0; \quad -(5\ \text{kN})(3\ \text{m}) + M_B = 0 \quad M_B = 15\ \text{kN}\cdot\text{m} \qquad \textit{Ans.}$$

Segment AC

$$\pm\rightarrow \Sigma F_x = 0; \quad N_C = 0 \qquad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \quad 5\ \text{kN} - 6\ \text{kN} - V_C = 0 \quad V_C = -1\ \text{kN} \qquad \textit{Ans.}$$

$$\circlearrowright +\Sigma M_C = 0; \quad -(5\ \text{kN})(3\ \text{m}) + M_C = 0 \quad M_C = 15\ \text{kN}\cdot\text{m} \qquad \textit{Ans.}$$

NOTE: The negative sign indicates that $\mathbf{V}_C$ acts in the opposite sense to that shown on the free-body diagram. Also, the moment arm for the 5-kN force in both cases is approximately 3 m since B and C are "almost" coincident.

Determine the normal force, shear force, and bending moment at C of the beam in Fig. 7–5*a*.

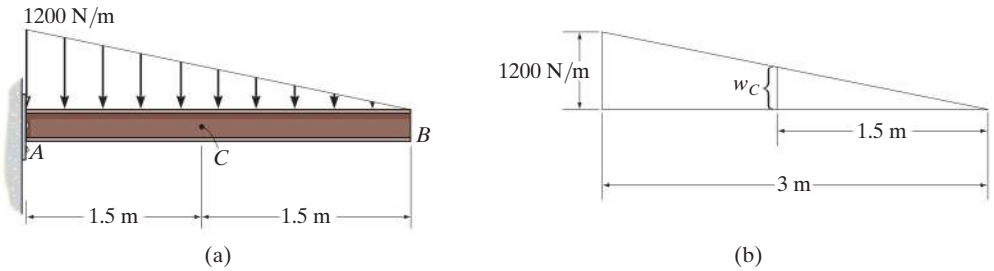
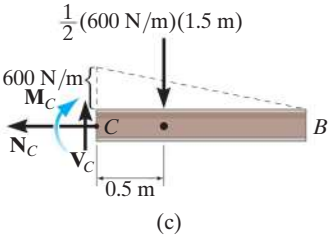


Fig. 7–5

SOLUTION

Free-Body Diagram. It is not necessary to find the support reactions at A since segment BC of the beam can be used to determine the internal loadings at C. The intensity of the triangular distributed load at C is determined using similar triangles from the geometry shown in Fig. 7–5*b*, i.e.,



$$w_C = (1200\ \text{N/m})\left(\frac{1.5\ \text{m}}{3\ \text{m}}\right) = 600\ \text{N/m}$$

The distributed load acting on segment BC can now be replaced by its resultant force, and its location is indicated on the free-body diagram, Fig. 7–5*c*.

Equations of Equilibrium.

$$\stackrel{+}{\rightarrow}\Sigma F_x = 0; \qquad N_C = 0 \qquad \textit{Ans.}$$

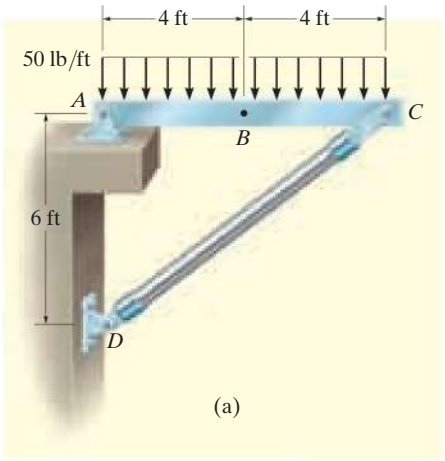
$$+\uparrow \Sigma F_y = 0; \qquad V_C - \tfrac{1}{2}(600\ \text{N/m})(1.5\ \text{m}) = 0$$

$$V_C = 450\ \text{N} \qquad \textit{Ans.}$$

$$\circlearrowleft + \Sigma M_C = 0; \qquad -M_C - \tfrac{1}{2}(600\ \text{N/m})(1.5\ \text{m})(0.5\ \text{m}) = 0$$

$$M_C = -225\ \text{N} \qquad \textit{Ans.}$$

The negative sign indicates that $\mathbf{M}_C$ acts in the opposite sense to that shown on the free-body diagram.



Determine the normal force, shear force, and bending moment acting at point B of the two-member frame shown in Fig. 7–6*a*.

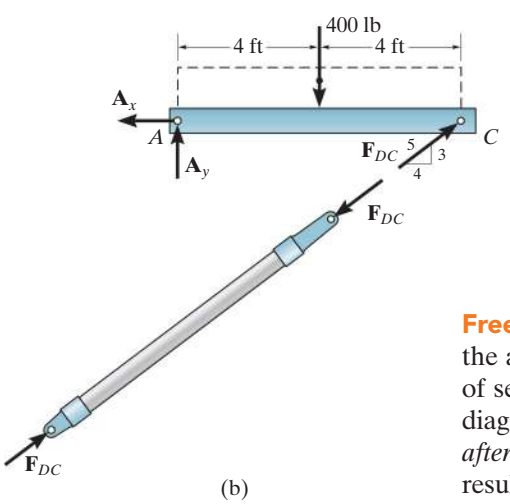
SOLUTION

Support Reactions. A free-body diagram of each member is shown in Fig. 7–6*b*. Since CD is a two-force member, the equations of equilibrium need to be applied only to member AC.

$$\zeta + \Sigma M_A = 0; \quad -400\text{ lb}(4\text{ ft}) + \left(\tfrac{3}{5}\right) F_{DC}(8\text{ ft}) = 0 \qquad F_{DC} = 333.3\text{ lb}$$

$$\pm \Sigma F_x = 0; \quad -A_x + \left(\tfrac{4}{5}\right)(333.3\text{ lb}) = 0 \qquad A_x = 266.7\text{ lb}$$

$$+\uparrow \Sigma F_y = 0; \quad A_y - 400\text{ lb} + \left(\tfrac{3}{5}\right)(333.3\text{ lb}) = 0 \qquad A_y = 200\text{ lb}$$



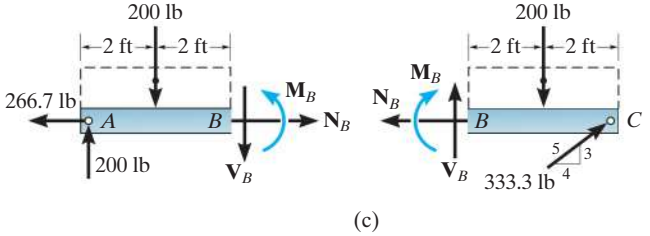


Fig. 7–6

Free-Body Diagrams. Passing an imaginary section perpendicular to the axis of member AC through point B yields the free-body diagrams of segments AB and BC shown in Fig. 7–6*c*. When constructing these diagrams it is important to keep the distributed loading where it is until *after the section is made*. Only then can it be replaced by a single resultant force.

Equations of Equilibrium. Applying the equations of equilibrium to segment AB, we have

$$\pm \Sigma F_x = 0; \quad N_B - 266.7\text{ lb} = 0 \qquad N_B = 267\text{ lb} \qquad \textit{Ans.}$$

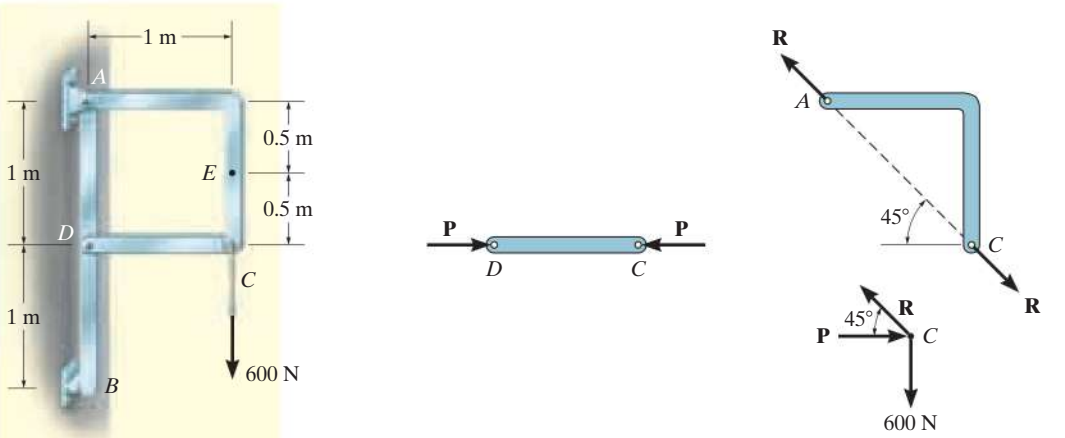
$$+\uparrow \Sigma F_y = 0; \quad 200\text{ lb} - 200\text{ lb} - V_B = 0 \qquad V_B = 0 \qquad \textit{Ans.}$$

$$\zeta + \Sigma M_B = 0; \quad M_B - 200\text{ lb}(4\text{ ft}) + 200\text{ lb}(2\text{ ft}) = 0$$

$$M_B = 400\text{ lb}\cdot\text{ft} \qquad \textit{Ans.}$$

NOTE: As an exercise, try to obtain these same results using segment BC.

Determine the normal force, shear force, and bending moment acting at point E of the frame loaded as shown in Fig. 7–7*a*.



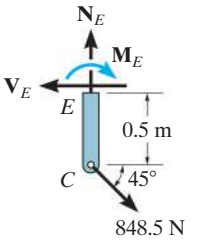
SOLUTION

Support Reactions. By inspection, members AC and CD are two-force members, Fig. 7–7*b*. In order to determine the internal loadings at E, we must first determine the force **R** acting at the end of member AC. To obtain it, we will analyze the equilibrium of the pin at C.

Summing forces in the vertical direction on the pin, Fig. 7–7*b*, we have

$$+\uparrow \Sigma F_y = 0; \qquad R \sin 45° - 600 \text{ N} = 0 \qquad R = 848.5 \text{ N}$$

Free-Body Diagram. The free-body diagram of segment CE is shown in Fig. 7–7*c*.



(c)

Fig. 7–7

Equations of Equilibrium.

$$\pm \Sigma F_x = 0; \qquad 848.5 \cos 45° \text{ N} - V_E = 0 \qquad V_E = 600 \text{ N} \quad \textit{Ans.}$$

$$+\uparrow \Sigma F_y = 0; \qquad -848.5 \sin 45° \text{ N} + N_E = 0 \qquad N_E = 600 \text{ N} \quad \textit{Ans.}$$

$$\circlearrowleft + \Sigma M_E = 0; \qquad 848.5 \cos 45° \text{ N}(0.5 \text{ m}) - M_E = 0$$

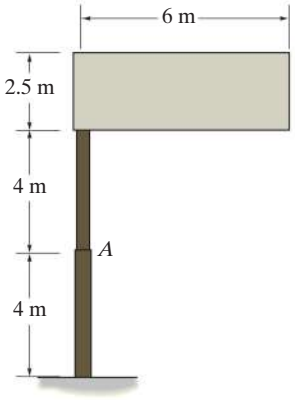
$$M_E = 300 \text{ N} \cdot \text{m} \quad \textit{Ans.}$$

NOTE: These results indicate a poor design. Member AC should be *straight* (from A to C) so that bending within the member is *eliminated*. If AC were straight then the internal force would only create tension in the member.

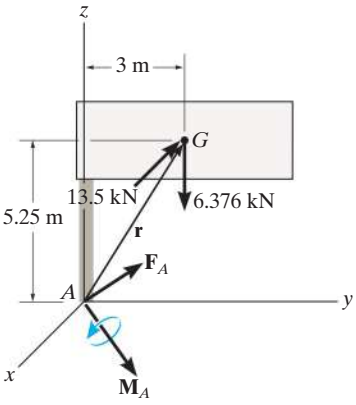


(© Russell C. Hibbeler)

(a)



(b)



(c)

Fig. 7–8

The uniform sign shown in Fig. 7–8*a* has a mass of 650 kg and is supported on the fixed column. Design codes indicate that the expected maximum uniform wind loading that will occur in the area where it is located is 900 Pa. Determine the internal loadings at *A*.

SOLUTION

The idealized model for the sign is shown in Fig. 7–8*b*. Here the necessary dimensions are indicated. We can consider the free-body diagram of a section above point *A* since it does not involve the support reactions.

Free-Body Diagram. The sign has a weight of $W = 650(9.81)\,\text{N} = 6.376\,\text{kN}$, and the wind creates a resultant force of $F_w = 900\,\text{N/m}^2(6\,\text{m})(2.5\,\text{m}) = 13.5\,\text{kN}$, which acts perpendicular to the face of the sign. These loadings are shown on the free-body diagram, Fig. 7–8*c*.

Equations of Equilibrium. Since the problem is three dimensional, a vector analysis will be used.

$$\Sigma\mathbf{F} = \mathbf{0}; \qquad \mathbf{F}_A - 13.5\mathbf{i} - 6.376\mathbf{k} = \mathbf{0}$$

$$\mathbf{F}_A = \{13.5\mathbf{i} + 6.38\mathbf{k}\}\ \text{kN} \qquad \textit{Ans.}$$

$$\Sigma\mathbf{M}_A = \mathbf{0}; \qquad \mathbf{M}_A + \mathbf{r} \times (\mathbf{F}_w + \mathbf{W}) = \mathbf{0}$$

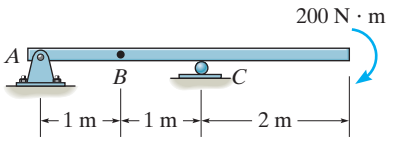
$$\mathbf{M}_A + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 3 & 5.25 \\ -13.5 & 0 & -6.376 \end{vmatrix} = \mathbf{0}$$

$$\mathbf{M}_A = \{19.1\mathbf{i} + 70.9\mathbf{j} - 40.5\mathbf{k}\}\ \text{kN}\cdot\text{m} \qquad \textit{Ans.}$$

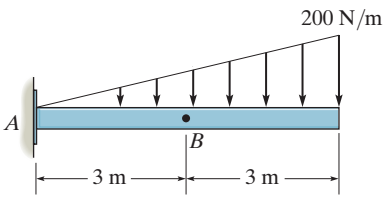
NOTE: Here $\mathbf{F}_{A_z} = \{6.38\mathbf{k}\}$ kN represents the normal force, whereas $\mathbf{F}_{A_x} = \{13.5\mathbf{i}\}$ kN is the shear force. Also, the torsional moment is $\mathbf{M}_{A_z} = \{-40.5\mathbf{k}\}\ \text{kN}\cdot\text{m}$, and the bending moment is determined from its components $\mathbf{M}_{A_x} = \{19.1\mathbf{i}\}\ \text{kN}\cdot\text{m}$ and $\mathbf{M}_{A_y} = \{70.9\mathbf{j}\}\ \text{kN}\cdot\text{m}$; i.e., $(M_b)_A = \sqrt{(M_A)_x^2 + (M_A)_y^2} = 73.4\ \text{kN}\cdot\text{m}$.

PRELIMINARY PROBLEMS

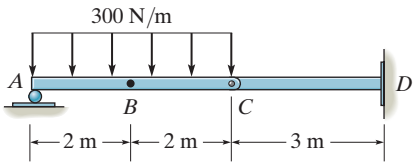
P7–1. In each case, calculate the reaction at A and then draw the free-body diagram of segment AB of the beam in order to determine the internal loading at B.



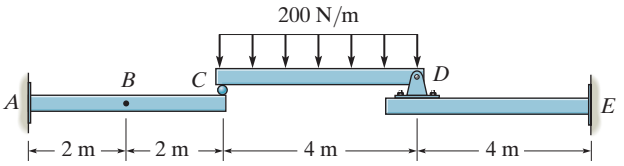
(a)



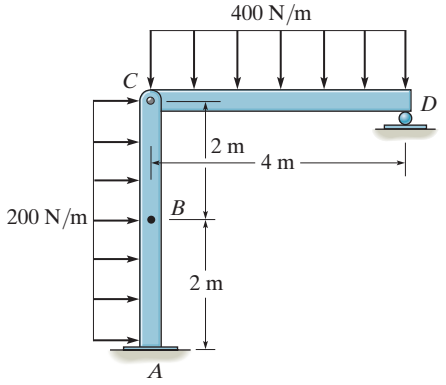
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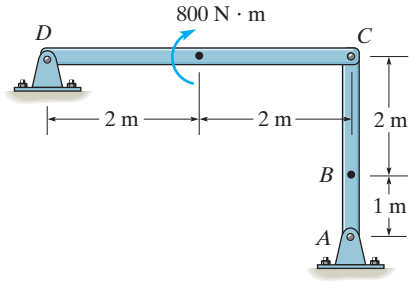
(c)



(d)



(e)



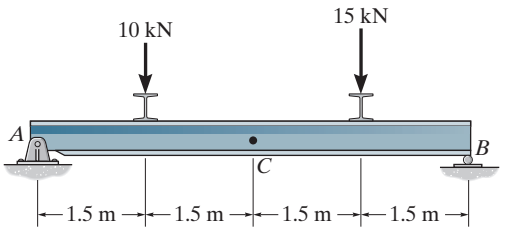
(f)

Prob. P7–1

FUNDAMENTAL PROBLEMS

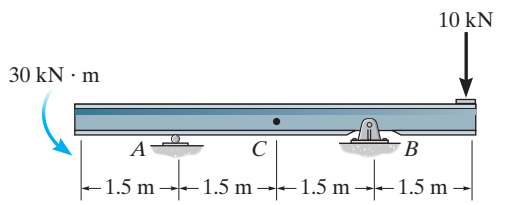
All problem solutions must include FBDs.

F7–1. Determine the normal force, shear force, and moment at point *C*.



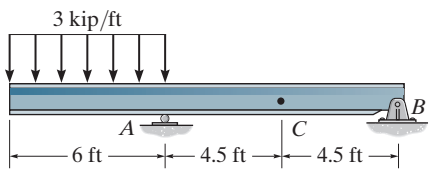
Prob. F7–1

F7–2. Determine the normal force, shear force, and moment at point *C*.



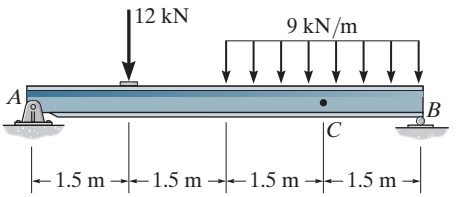
Prob. F7–2

F7–3. Determine the normal force, shear force, and moment at point *C*.



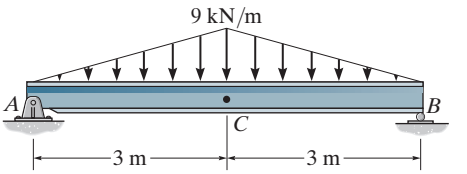
Prob. F7–3

F7–4. Determine the normal force, shear force, and moment at point *C*.



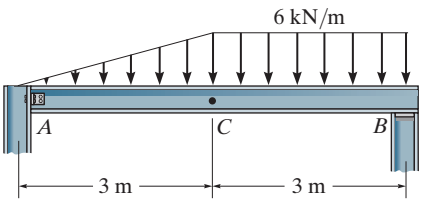
Prob. F7–4

F7–5. Determine the normal force, shear force, and moment at point *C*.



Prob. F7–5

F7–6. Determine the normal force, shear force, and moment at point *C*. Assume *A* is pinned and *B* is a roller.

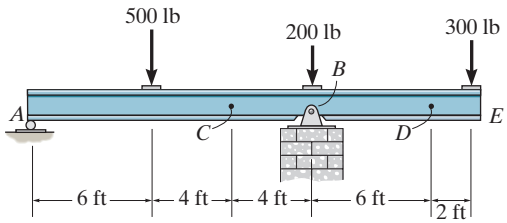


Prob. F7–6

PROBLEMS

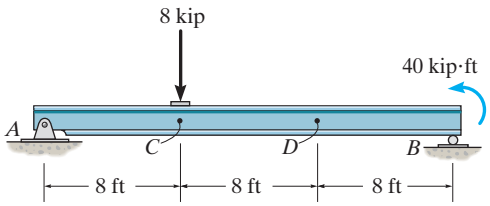
All problem solutions must include FBDs.

7–1. Determine the shear force and moment at points *C* and *D*.



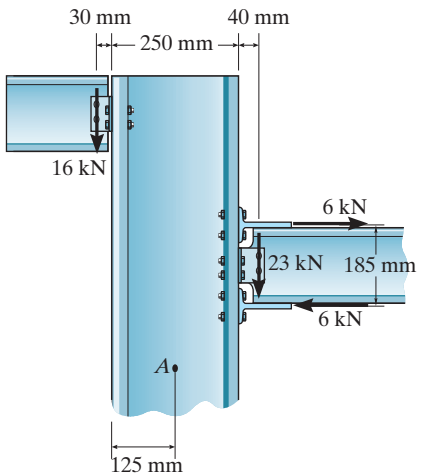
Prob. 7–1

7–2. Determine the internal normal force and shear force, and the bending moment in the beam at points *C* and *D*. Assume the support at *B* is a roller. Point *C* is located just to the right of the 8-kip load.



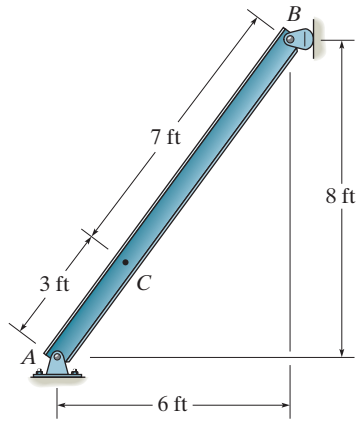
Prob. 7–2

7–3. Two beams are attached to the column such that structural connections transmit the loads shown. Determine the internal normal force, shear force, and moment acting in the column at a section passing horizontally through point *A*.



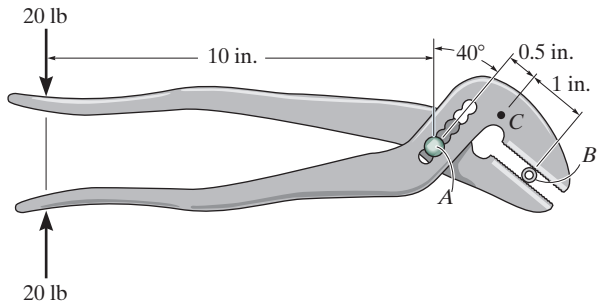
Prob. 7–3

***7–4.** The beam weighs 280 lb/ft. Determine the internal normal force, shear force, and moment at point *C*.



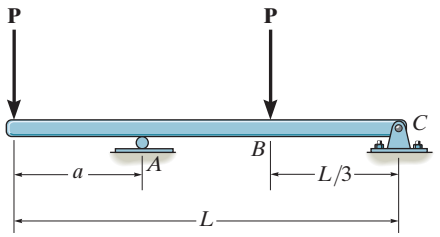
Prob. 7–4

7–5. The pliers are used to grip the tube at *B*. If a force of 20 lb is applied to the handles, determine the internal shear force and moment a point *C*. Assume the jaws of the pliers exert only normal forces on the tube.



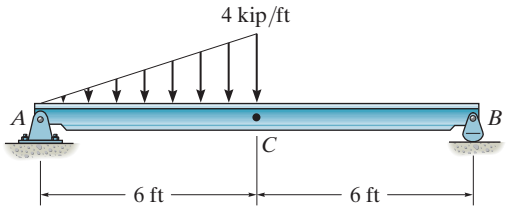
Prob. 7–5

7–6. Determine the distance a as a fraction of the beam's length L for locating the roller support so that the moment in the beam at B is zero.



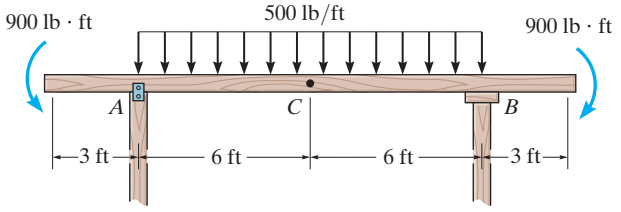
Prob. 7–6

7–7. Determine the internal shear force and moment acting at point C in the beam.



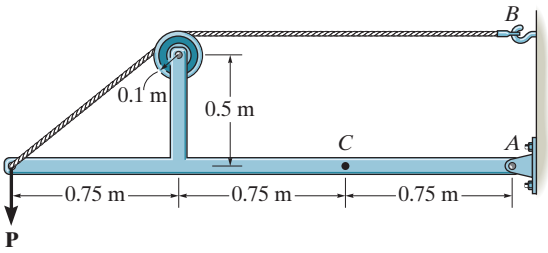
Prob. 7–7

***7–8.** Determine the internal shear force and moment acting at point C in the beam.



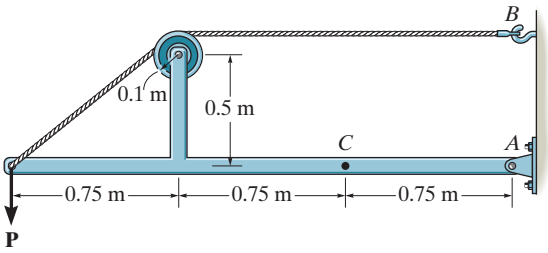
Prob. 7–8

7–9. Determine the normal force, shear force, and moment at a section passing through point C. Take $P = 8$ kN.



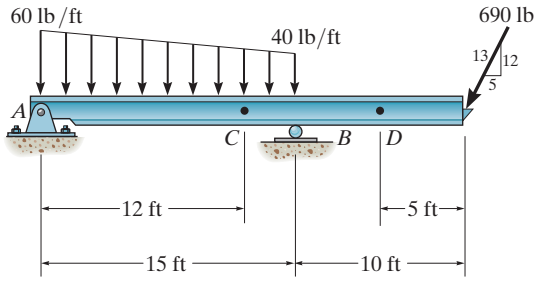
Prob. 7–9

7–10. The cable will fail when subjected to a tension of 2 kN. Determine the largest vertical load P the frame will support and calculate the internal normal force, shear force, and moment at a section passing through point C for this loading.



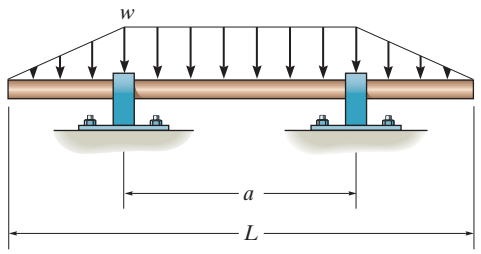
Prob. 7–10

7–11. Determine the internal normal force, shear force, and moment at points C and D of the beam.



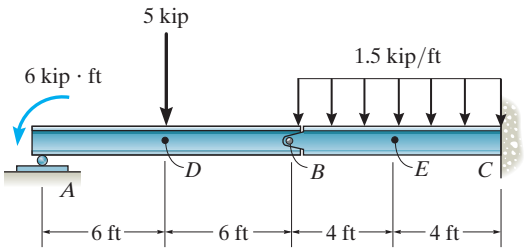
Prob. 7–11

*7–12. Determine the distance a between the bearings in terms of the shaft's length L so that the moment in the *symmetric* shaft is zero at its center.



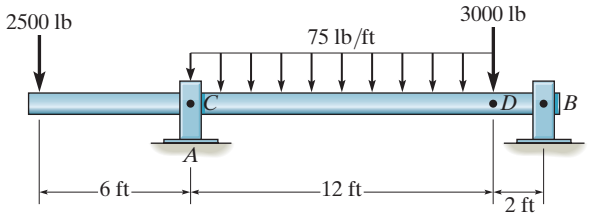
Prob. 7–12

7–13. Determine the internal normal force, shear force, and moment in the beam at sections passing through points D and E. Point D is located just to the left of the 5-kip load.



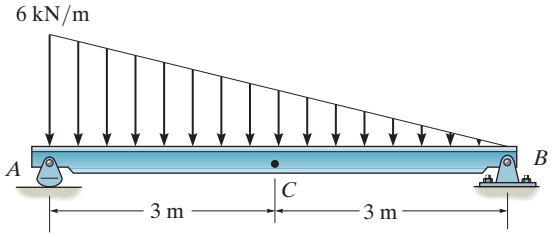
Prob. 7–13

7–14. The shaft is supported by a journal bearing at A and a thrust bearing at B. Determine the normal force, shear force, and moment at a section passing through (a) point C, which is just to the right of the bearing at A, and (b) point D, which is just to the left of the 3000-lb force.



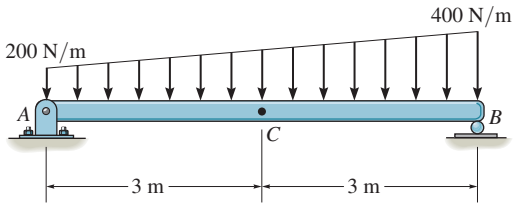
Prob. 7–14

7–15. Determine the internal normal force, shear force, and moment at point C.



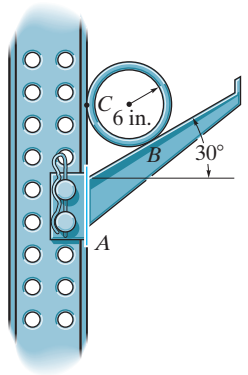
Prob. 7–15

*7–16. Determine the internal normal force, shear force, and moment at point C of the beam.



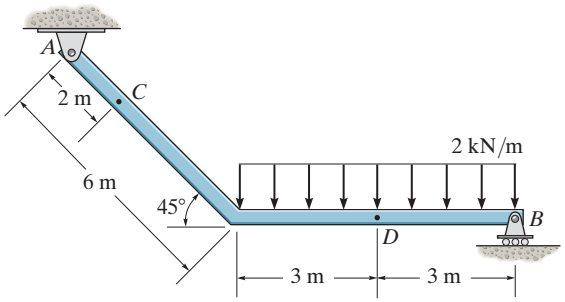
Prob. 7–16

7–17. The cantilevered rack is used to support each end of a smooth pipe that has a total weight of 300 lb. Determine the normal force, shear force, and moment that act in the arm at its fixed support A along a vertical section.



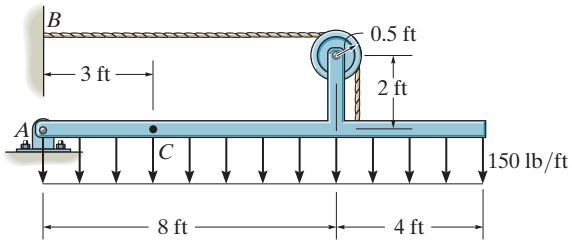
Prob. 7–17

7–18. Determine the internal normal force, shear force, and the moment at points *C* and *D*.



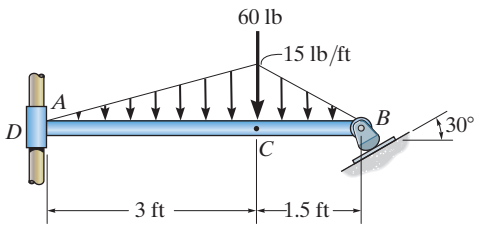
Prob. 7–18

7–19. Determine the internal normal force, shear force, and moment at point *C*.



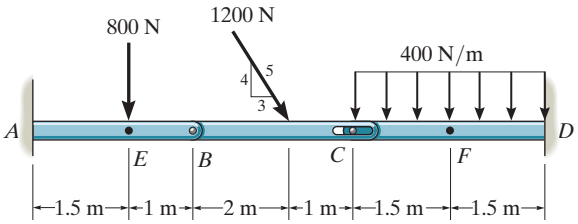
Prob. 7–19

***7–20.** Rod *AB* is fixed to a smooth collar *D*, which slides freely along the vertical guide. Determine the internal normal force, shear force, and moment at point *C*, which is located just to the left of the 60-lb concentrated load.



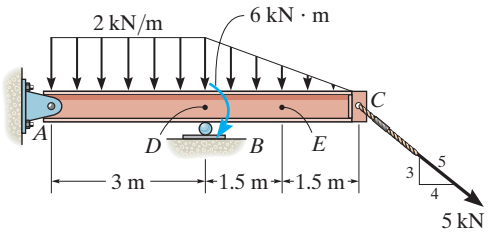
Prob. 7–20

7–21. Determine the internal normal force, shear force, and moment at points *E* and *F* of the compound beam. Point *E* is located just to the left of 800 N force.



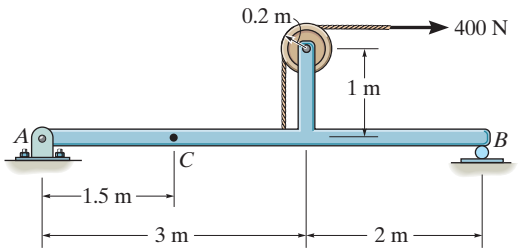
Prob. 7–21

7–22. Determine the internal normal force, shear force, and moment at points *D* and *E* in the overhang beam. Point *D* is located just to the left of the roller support at *B*, where the couple moment acts.



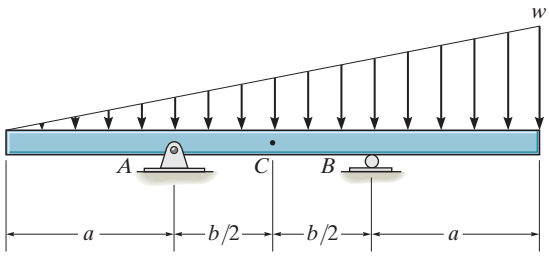
Prob. 7–22

7–23. Determine the internal normal force, shear force, and moment at point *C*.



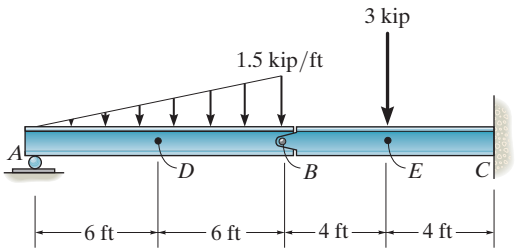
Prob. 7–23

***7–24.** Determine the ratio of a/b for which the shear force will be zero at the midpoint C of the beam.



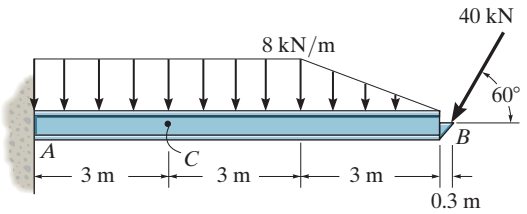
Prob. 7–24

7–25. Determine the normal force, shear force, and moment in the beam at sections passing through points D and E. Point E is just to the right of the 3-kip load.



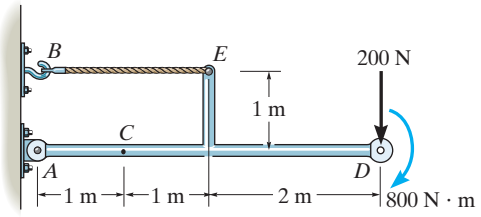
Prob. 7–25

7–26. Determine the internal normal force, shear force, and bending moment at point C.



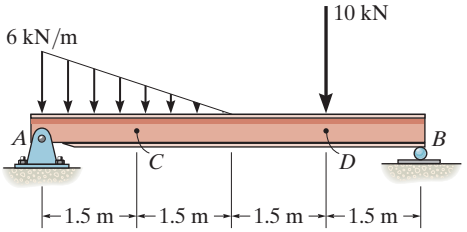
Prob. 7–26

7–27. Determine the internal normal force, shear force, and moment at point C.



Prob. 7–27

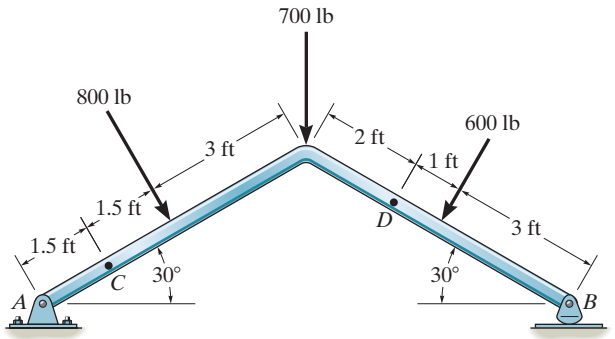
***7–28.** Determine the internal normal force, shear force, and moment at points C and D in the simply supported beam. Point D is located just to the left of the 10-kN concentrated load.



Prob. 7–28

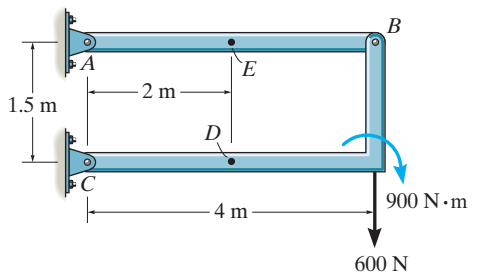
7–29. Determine the normal force, shear force, and moment acting at a section passing through point *C*.

7–30. Determine the normal force, shear force, and moment acting at a section passing through point *D*.



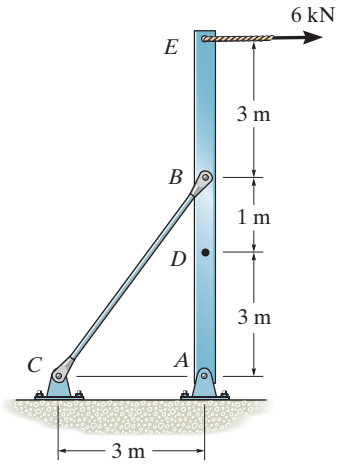
Probs. 7–29/30

7–31. Determine the internal normal force, shear force, and moment acting at points *D* and *E* of the frame.



Prob. 7–31

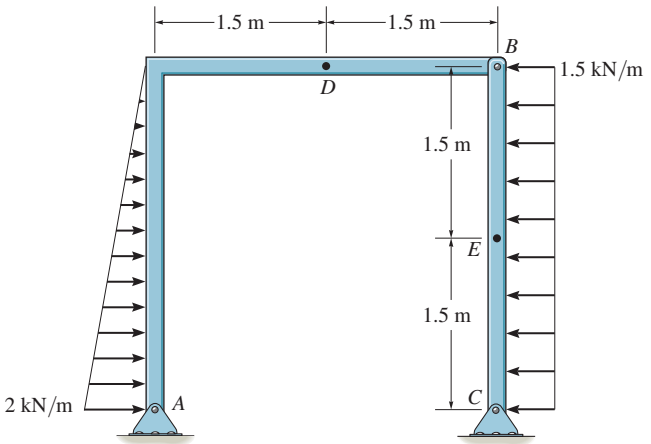
***7–32.** Determine the internal normal force, shear force, and moment at point *D*.



Prob. 7–32

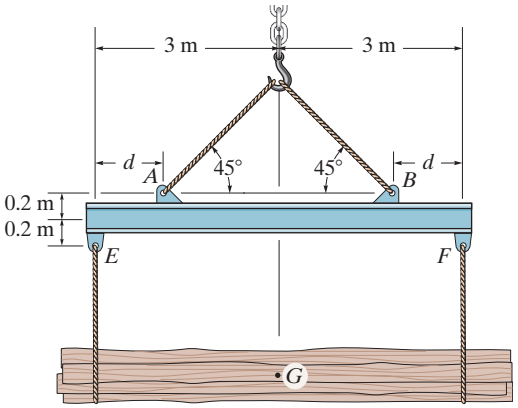
7–33. Determine the internal normal force, shear force, and moment at point *D* of the two-member frame.

7–34. Determine the internal normal force, shear force, and moment at point *E*.



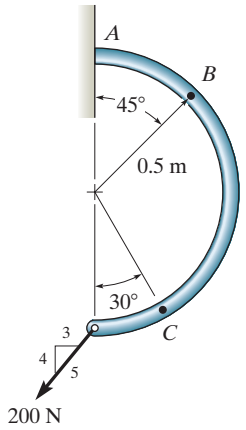
Probs. 7–33/34

7–35. The strongback or lifting beam is used for materials handling. If the suspended load has a weight of 2 kN and a center of gravity of G, determine the placement d of the padeyes on the top of the beam so that there is no moment developed within the length AB of the beam. The lifting bridle has two legs that are positioned at 45°, as shown.



Prob. 7–35

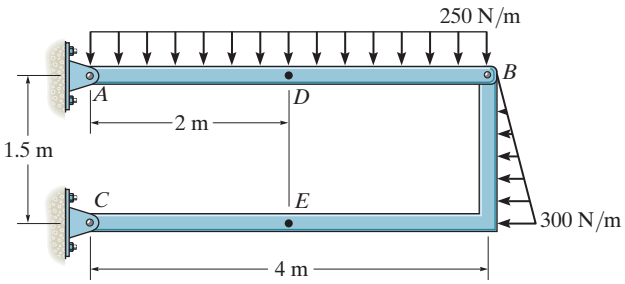
***7–36.** Determine the internal normal force, shear force, and moment acting at points B and C on the curved rod.



Prob. 7–36

7–37. Determine the internal normal force, shear force, and moment at point D of the two-member frame.

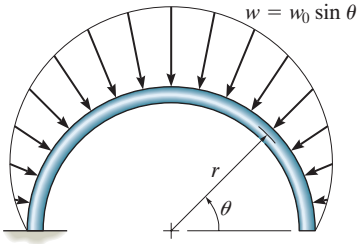
7–38. Determine the internal normal force, shear force, and moment at point E of the two-member frame.



Probs. 7–37/38

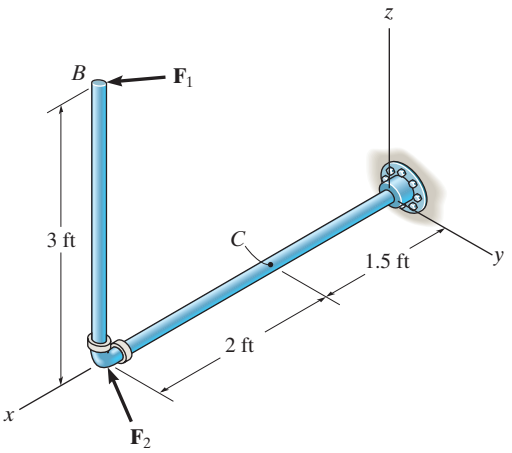
7–39. The distributed loading $w = w_0 \sin \theta$, measured per unit length, acts on the curved rod. Determine the internal normal force, shear force, and moment in the rod at $\theta = 45°$.

***7–40.** Solve Prob. 7–39 for $\theta = 120°$.



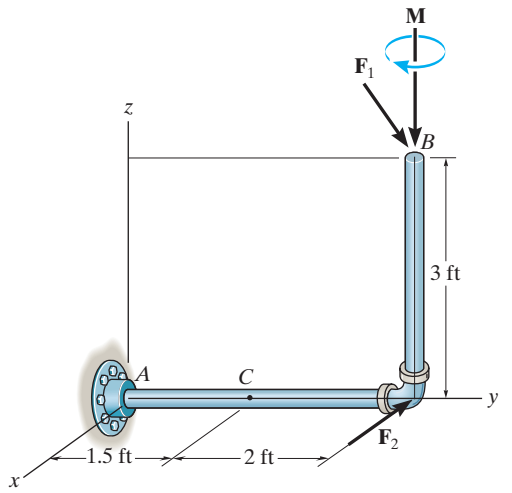
Probs. 7–39/40

7–41. Determine the *x*, *y*, *z* components of force and moment at point *C* in the pipe assembly. Neglect the weight of the pipe. Take $\mathbf{F}_1 = \{350\mathbf{i} - 400\mathbf{j}\}$ lb and $\mathbf{F}_2 = \{-300\mathbf{j} + 150\mathbf{k}\}$ lb.



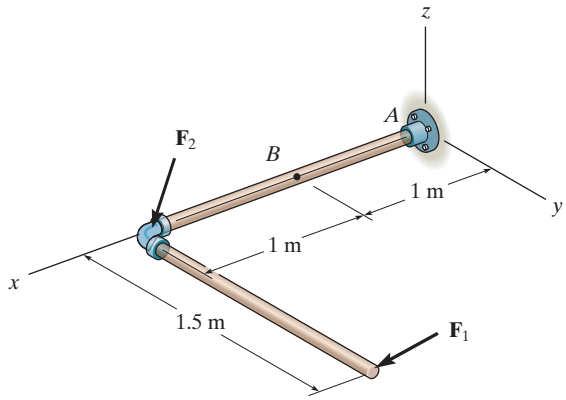
Prob. 7–41

7–42. Determine the *x*, *y*, *z* components of force and moment at point *C* in the pipe assembly. Neglect the weight of the pipe. The load acting at (0, 3.5 ft, 3 ft) is $\mathbf{F}_1 = \{-24\mathbf{i} - 10\mathbf{k}\}$ lb and $\mathbf{M} = \{-30\mathbf{k}\}$ lb · ft and at point (0, 3.5 ft, 0) $\mathbf{F}_2 = \{-80\mathbf{i}\}$ lb.



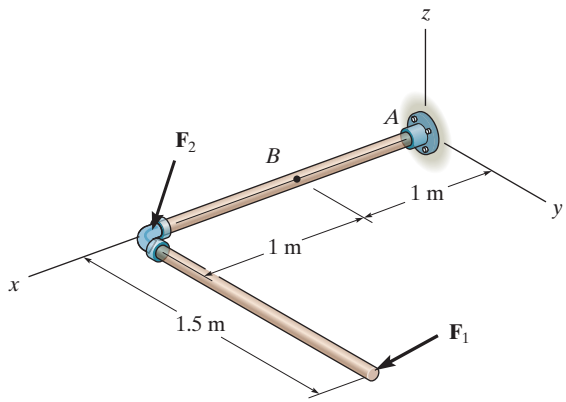
Prob. 7–42

7–43. Determine the *x*, *y*, *z* components of internal loading at a section passing through point *B* in the pipe assembly. Neglect the weight of the pipe. Take $\mathbf{F}_1 = \{200\mathbf{i} - 100\mathbf{j} - 400\mathbf{k}\}$ N and $\mathbf{F}_2 = \{300\mathbf{i} - 500\mathbf{k}\}$ N.



Prob. 7–43

***7–44.** Determine the *x*, *y*, *z* components of internal loading at a section passing through point *B* in the pipe assembly. Neglect the weight of the pipe. Take $\mathbf{F}_1 = \{100\mathbf{i} - 200\mathbf{j} - 300\mathbf{k}\}$ N and $\mathbf{F}_2 = \{100\mathbf{i} + 500\mathbf{j}\}$ N.



Prob. 7–44

*7.2 Shear and Moment Equations and Diagrams

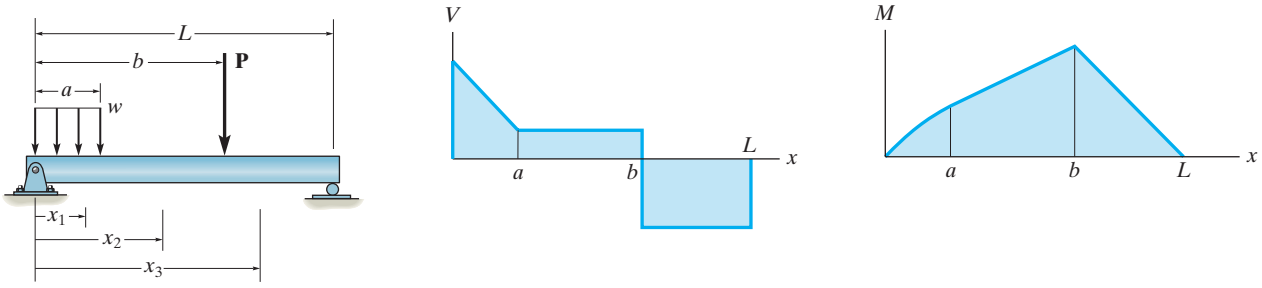
Beams are structural members designed to support loadings applied perpendicular to their axes. In general, they are long and straight and have a constant cross-sectional area. They are often classified as to how they are supported. For example, a ***simply supported beam*** is pinned at one end and roller supported at the other, as in Fig. 7–9*a*, whereas a ***cantilevered beam*** is fixed at one end and free at the other. The actual design of a beam requires a detailed knowledge of the *variation* of the internal shear force V and bending moment M acting at *each point* along the axis of the beam.*

These *variations* of V and M along the beam's axis can be obtained by using the method of sections discussed in Sec. 7.1. In this case, however, it is necessary to section the beam at an arbitrary distance x from one end and then apply the equations of equilibrium to the segment having the length x. Doing this we can then obtain V and M as functions of x.

In general, the internal shear and bending-moment functions will be discontinuous, or their slopes will be discontinuous, at points where a distributed load changes or where concentrated forces or couple moments are applied. Because of this, these functions must be determined for *each segment* of the beam located between any two discontinuities of loading. For example, segments having lengths x_1, x_2, and x_3 will have to be used to describe the variation of V and M along the length of the beam in Fig. 7–9*a*. These functions will be valid *only* within regions from 0 to a for x_1, from a to b for x_2, and from b to L for x_3. If the resulting functions of x are plotted, the graphs are termed the ***shear diagram*** and ***bending-moment diagram***, Fig. 7–9*b* and Fig. 7–9*c*, respectively.



To save on material and thereby produce an efficient design, these beams, also called girders, have been tapered, since the internal moment in the beam will be larger at the supports, or piers, than at the center of the span. (© Russell C. Hibbeler)



(a) (b) (c)

Fig. 7–9

*The internal normal force is not considered for two reasons. In most cases, the loads applied to a beam act perpendicular to the beam's axis and hence produce only an internal shear force and bending moment. And for design purposes, the beam's resistance to shear, and particularly to bending, is more important than its ability to resist a normal force.

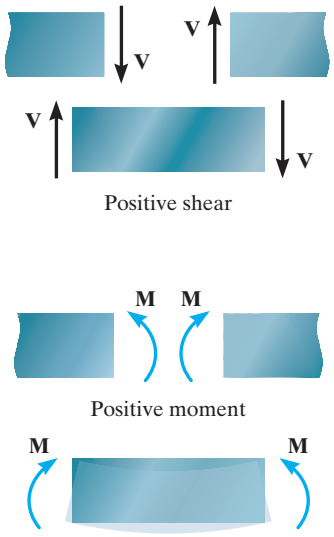


Fig. 7–10

Important Points

- Shear and moment diagrams for a beam provide graphical descriptions of how the internal shear and moment vary throughout the beam's length.
- To obtain these diagrams, the method of sections is used to determine V and M as functions of x. These results are then plotted. If the load on the beam suddenly changes, then regions between each load must be selected to obtain each function of x.

Procedure for Analysis

The shear and bending-moment diagrams for a beam can be constructed using the following procedure.

Support Reactions.

- Determine all the reactive forces and couple moments acting on the beam and resolve all the forces into components acting perpendicular and parallel to the beam's axis.

Shear and Moment Functions.

- Specify separate coordinates x having an origin at the beam's left end and extending to regions of the beam *between* concentrated forces and/or couple moments, or where the distributed loading is continuous.
- Section the beam at each distance x and draw the free-body diagram of one of the segments. Be sure **V** and **M** are shown acting in their *positive sense*, in accordance with the sign convention given in Fig. 7–10.
- The shear V is obtained by summing forces perpendicular to the beam's axis, and the moment M is obtained by summing moments about the sectioned end of the segment.

Shear and Moment Diagrams.

- Plot the shear diagram (V versus x) and the moment diagram (M versus x). If computed values of the functions describing V and M are *positive*, the values are plotted above the x axis, whereas *negative* values are plotted below the x axis.



The shelving arms must be designed to resist the internal loading in the arms caused by the lumber. (© Russell C. Hibbeler)

Draw the shear and moment diagrams for the shaft shown in Fig. 7–11*a*. The support at *A* is a thrust bearing and the support at *C* is a journal bearing.

SOLUTION

Support Reactions. The support reactions are shown on the shaft's free-body diagram, Fig. 7–11*d*.

Shear and Moment Functions. The shaft is sectioned at an arbitrary distance x from point A, extending within the region AB, and the free-body diagram of the left segment is shown in Fig. 7–11*b*. The unknowns **V** and **M** are assumed to act in the *positive sense* on the right-hand face of the segment according to the established sign convention. Applying the equilibrium equations yields

$$+\uparrow \Sigma F_y = 0; \qquad V = 2.5 \text{ kN} \tag{1}$$

$$\zeta + \Sigma M = 0; \qquad M = 2.5x \text{ kN} \cdot \text{m} \tag{2}$$

A free-body diagram for a left segment of the shaft extending from A a distance x, within the region BC is shown in Fig. 7–11*c*. As always, **V** and **M** are shown acting in the positive sense. Hence,

$$+\uparrow \Sigma F_y = 0; \qquad 2.5 \text{ kN} - 5 \text{ kN} - V = 0$$

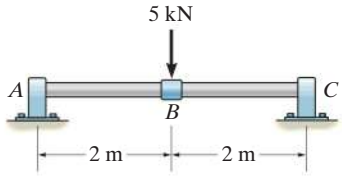
$$V = -2.5 \text{ kN} \tag{3}$$

$$\zeta + \Sigma M = 0; \qquad M + 5 \text{ kN}(x - 2 \text{ m}) - 2.5 \text{ kN}(x) = 0$$

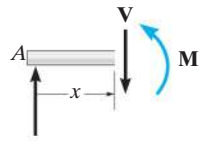
$$M = (10 - 2.5x) \text{ kN} \cdot \text{m} \tag{4}$$

Shear and Moment Diagrams. When Eqs. 1 through 4 are plotted within the regions in which they are valid, the shear and moment diagrams shown in Fig. 7–11*d* are obtained. The shear diagram indicates that the internal shear force is always 2.5 kN (positive) within segment AB. Just to the right of point B, the shear force changes sign and remains at a constant value of −2.5 kN for segment BC. The moment diagram starts at zero, increases linearly to point B at $x = 2$ m, where $M_{\max} = 2.5 \text{ kN}(2 \text{ m}) = 5 \text{ kN} \cdot \text{m}$, and thereafter decreases back to zero.

NOTE: It is seen in Fig. 7–11*d* that the graphs of the shear and moment diagrams "jump" or changes abruptly where the concentrated force acts, i.e., at points *A, B*, and *C*. For this reason, as stated earlier, it is necessary to express both the shear and moment functions separately for regions between concentrated loads. It should be realized, however, that all loading discontinuities are mathematical, arising from the *idealization of a concentrated force and couple moment*. Physically, loads are always applied over a finite area, and if the actual load variation could be accounted for, the shear and moment diagrams would then be continuous over the shaft's entire length.

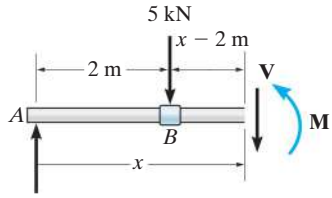


(a)



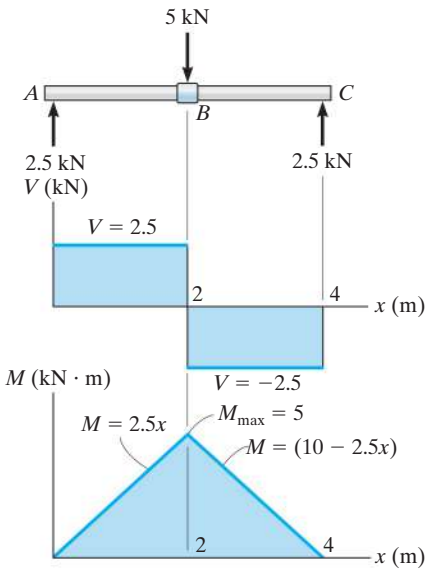
$0 \le x < 2$ m

(b)



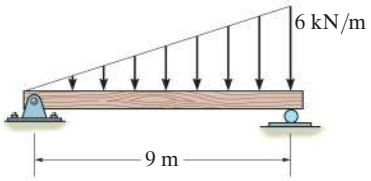
$2 \text{ m} < x \le 4$ m

(c)

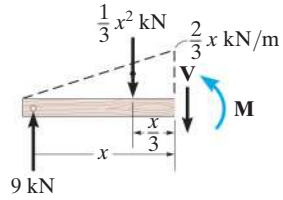


(d)

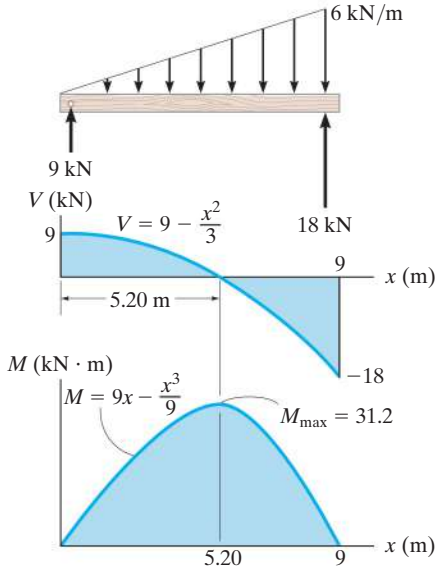
Fig. 7–11



(a)



(b)



(c)

Fig. 7–12

Draw the shear and moment diagrams for the beam shown in Fig. 7–12*a*.

SOLUTION

Support Reactions. The support reactions are shown on the beam's free-body diagram, Fig. 7–12*c*.

Shear and Moment Functions. A free-body diagram for a left segment of the beam having a length x is shown in Fig. 7–12*b*. Due to proportional triangles, the distributed loading acting at the end of this segment has an intensity of $w/x = 6/9$ or $w = (2/3)x$. It is replaced by a resultant force *after* the segment is isolated as a free-body diagram. The *magnitude* of the resultant force is equal to $\frac{1}{2}(x)\left(\frac{2}{3}x\right) = \frac{1}{3}x^2$. This force *acts through the centroid* of the distributed loading area, a distance $\frac{1}{3}x$ from the right end. Applying the two equations of equilibrium yields

$$+\uparrow \Sigma F_y = 0; \qquad 9 - \frac{1}{3}x^2 - V = 0$$

$$V = \left(9 - \frac{x^2}{3}\right) \text{kN} \tag{1}$$

$$\circlearrowleft +\Sigma M = 0; \qquad M + \frac{1}{3}x^2\left(\frac{x}{3}\right) - 9x = 0$$

$$M = \left(9x - \frac{x^3}{9}\right) \text{kN} \cdot \text{m} \tag{2}$$

Shear and Moment Diagrams. The shear and moment diagrams shown in Fig. 7–12*c* are obtained by plotting Eqs. 1 and 2.

The point of *zero shear* can be found using Eq. 1:

$$V = 9 - \frac{x^2}{3} = 0$$

$$x = 5.20 \text{ m}$$

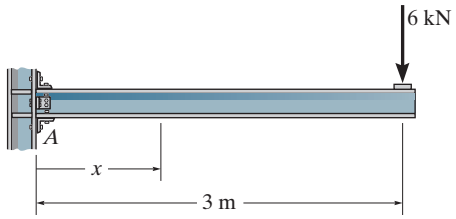
NOTE: It will be shown in Sec. 7.3 that this value of x happens to represent the point on the beam where the *maximum moment* occurs. Using Eq. 2, we have

$$M_{\max} = \left(9(5.20) - \frac{(5.20)^3}{9}\right) \text{kN} \cdot \text{m}$$

$$= 31.2 \text{ kN} \cdot \text{m}$$

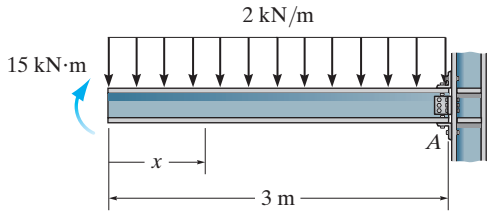
FUNDAMENTAL PROBLEMS

F7–7. Determine the shear and moment as a function of x, and then draw the shear and moment diagrams.



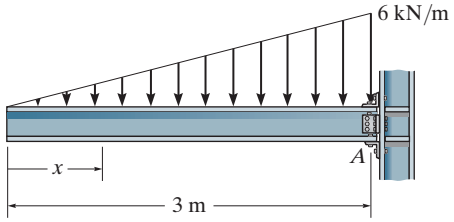
Prob. F7–7

F7–8. Determine the shear and moment as a function of x, and then draw the shear and moment diagrams.



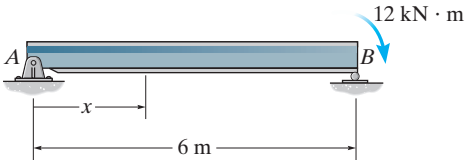
Prob. F7–8

F7–9. Determine the shear and moment as a function of x, and then draw the shear and moment diagrams.



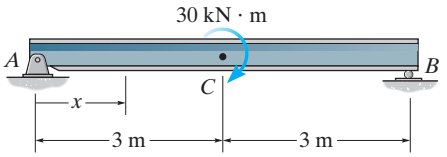
Prob. F7–9

F7–10. Determine the shear and moment as a function of x, and then draw the shear and moment diagrams.



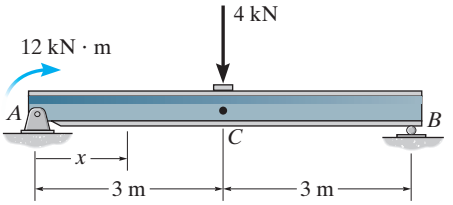
Prob. F7–10

F7–11. Determine the shear and moment as a function of x, where $0 \le x < 3$ m and 3 m $< x \le 6$ m, and then draw the shear and moment diagrams.



Prob. F7–11

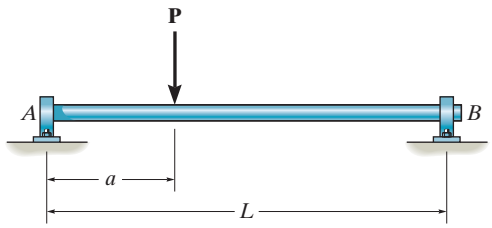
F7–12. Determine the shear and moment as a function of x, where $0 \le x < 3$ m and 3 m $< x \le 6$ m, and then draw the shear and moment diagrams.



Prob. F7–12

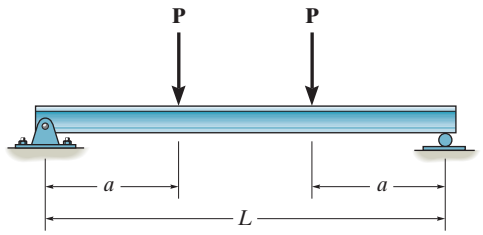
PROBLEMS

7–45. Draw the shear and moment diagrams for the shaft (a) in terms of the parameters shown; (b) set $P = 9$ kN, $a = 2$ m, $L = 6$ m. There is a thrust bearing at A and a journal bearing at B.



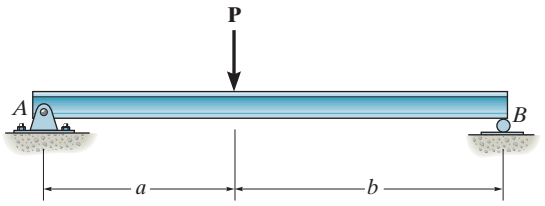
Prob. 7–45

7–46. Draw the shear and moment diagrams for the beam (a) in terms of the parameters shown; (b) set $P = 800$ lb, $a = 5$ ft, $L = 12$ ft.



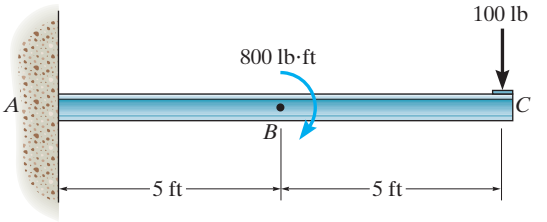
Prob. 7–46

7–47. Draw the shear and moment diagrams for the beam (a) in terms of the parameters shown; (b) set $P = 600$ lb, $a = 5$ ft, $b = 7$ ft.



Prob. 7–47

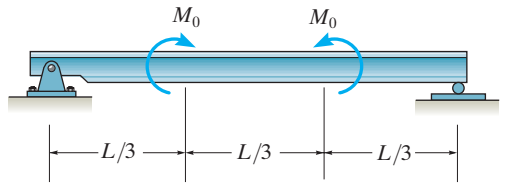
***7–48.** Draw the shear and moment diagrams for the cantilevered beam.



Prob. 7–48

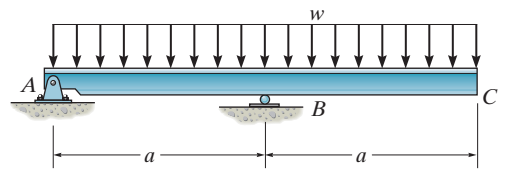
7–49. Draw the shear and moment diagrams of the beam (a) in terms of the parameters shown; (b) set $M_0 = 500$ N · m, $L = 8$ m.

7–50. If $L = 9$ m, the beam will fail when the maximum shear force is $V_{max} = 5$ kN or the maximum bending moment is $M_{max} = 2$ kN · m. Determine the magnitude M_0 of the largest couple moments it will support.



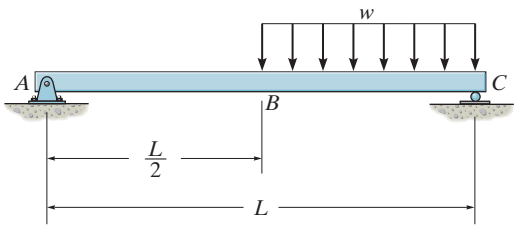
Probs. 7–49/50

7–51. Draw the shear and moment diagrams for the beam.



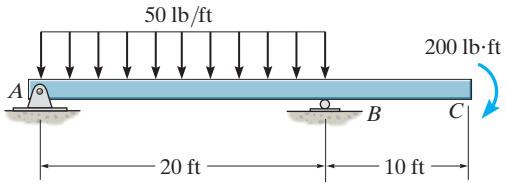
Prob. 7–51

*7–52. Draw the shear and moment diagrams for the beam.



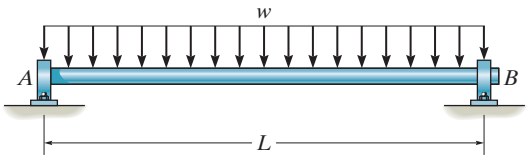
Prob. 7–52

7–53. Draw the shear and bending-moment diagrams for the beam.



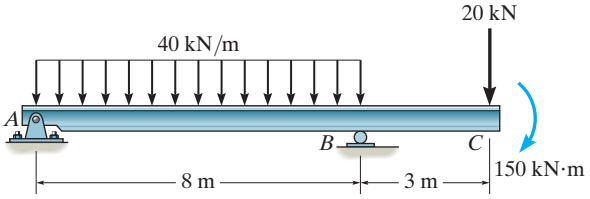
Prob. 7–53

7–54. The shaft is supported by a smooth thrust bearing at A and a smooth journal bearing at B. Draw the shear and moment diagrams for the shaft (a) in terms of the parameters shown; (b) set $w = 500$ lb/ft, $L = 10$ ft.



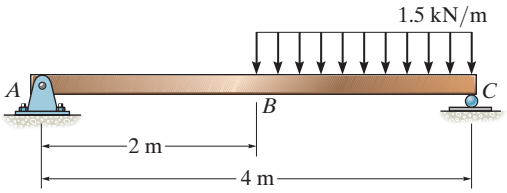
Prob. 7–54

7–55. Draw the shear and moment diagrams for the beam.



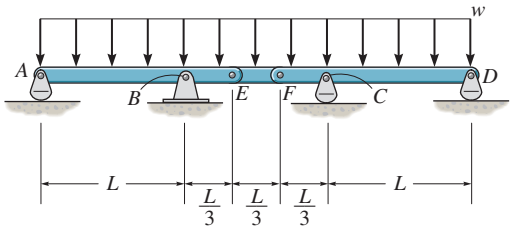
Prob. 7–55

*7–56. Draw the shear and moment diagrams for the beam.



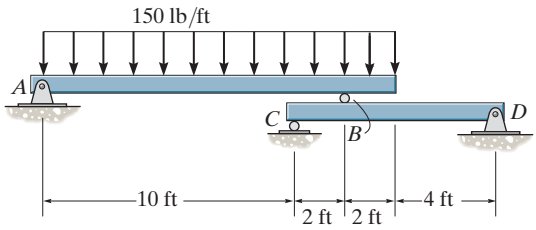
Prob. 7–56

7–57. Draw the shear and moment diagrams for the compound beam. The beam is pin connected at E and F.



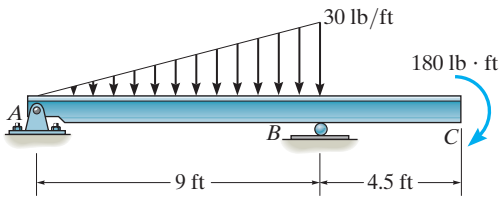
Prob. 7–57

7–58. Draw the shear and bending-moment diagrams for each of the two segments of the compound beam.



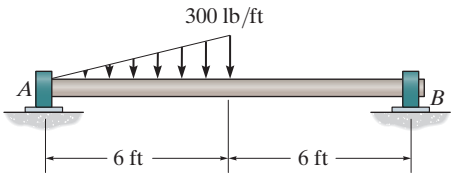
Prob. 7–58

7–59. Draw the shear and moment diagrams for the beam.



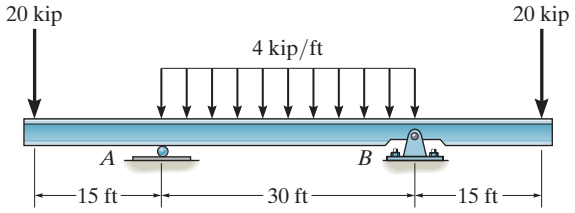
Prob. 7–59

*7–60. The shaft is supported by a smooth thrust bearing at A and a smooth journal bearing at B. Draw the shear and moment diagrams for the shaft.



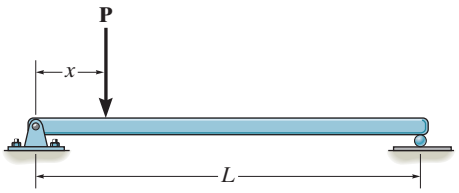
Prob. 7–60

7–61. Draw the shear and moment diagrams for the beam.



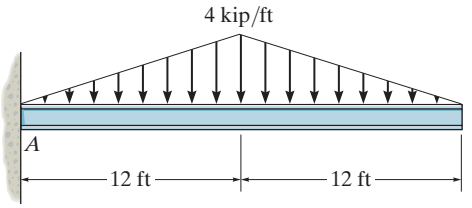
Prob. 7–61

7–62. The beam will fail when the maximum internal moment is M_{max}. Determine the position x of the concentrated force **P** and its smallest magnitude that will cause failure.



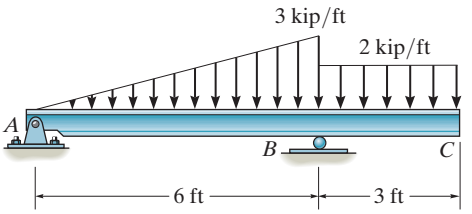
Prob. 7–62

7–63. Draw the shear and moment diagrams for the beam.



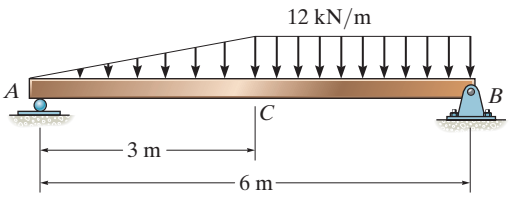
Prob. 7–63

*7–64. Draw the shear and moment diagrams for the beam.



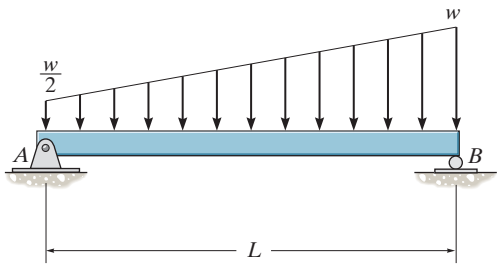
Prob. 7–64

7–65. Draw the shear and moment diagrams for the beam.



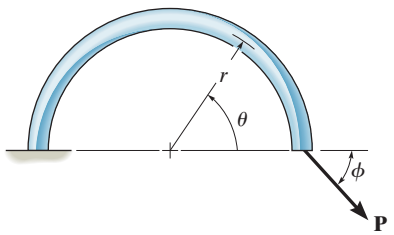
Prob. 7–65

7–66. Draw the shear and moment diagrams for the beam.



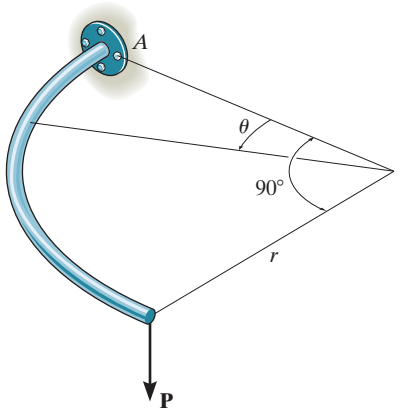
Prob. 7–66

7–67. Determine the internal normal force, shear force, and moment in the curved rod as a function of θ. The force **P** acts at the constant angle ϕ.



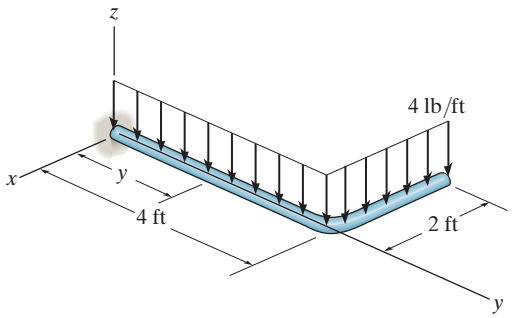
Prob. 7–67

***7–68.** The quarter circular rod lies in the horizontal plane and supports a vertical force **P** at its end. Determine the magnitudes of the components of the internal shear force, moment, and torque acting in the rod as a function of the angle θ.



Prob. 7–68

7–69. Express the internal shear and moment components acting in the rod as a function of y, where $0 \leq y \leq 4$ ft.



Prob. 7–69

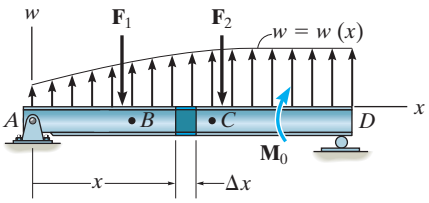
*7.3 Relations between Distributed Load, Shear, and Moment



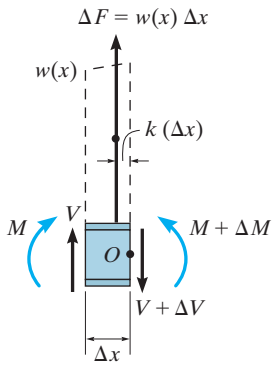
In order to design the beam used to support these power lines, it is important to first draw the shear and moment diagrams for the beam. (© Russell C. Hibbeler)

If a beam is subjected to several concentrated forces, couple moments, and distributed loads, the method of constructing the shear and bending-moment diagrams discussed in Sec. 7.2 may become quite tedious. In this section a simpler method for constructing these diagrams is discussed—a method based on differential relations that exist between the load, shear, and bending moment.

Distributed Load. Consider the beam AD shown in Fig. 7–13*a*, which is subjected to an arbitrary load $w = w(x)$ and a series of concentrated forces and couple moments. In the following discussion, the *distributed load* will be considered *positive* when the *loading acts upward* as shown. A free-body diagram for a small segment of the beam having a length Δx is chosen at a point x along the beam which is *not* subjected to a concentrated force or couple moment, Fig. 7–13*b*. Hence any results obtained will not apply at these points of concentrated loading. The internal shear force and bending moment shown on the free-body diagram are assumed to act in the *positive sense* according to the established sign convention. Note that both the shear force and moment acting on the right-hand face must be increased by a small, finite amount in order to keep the segment in equilibrium. The distributed loading has been replaced by a resultant force $\Delta F = w(x)\,\Delta x$ that acts at a fractional distance $k(\Delta x)$ from the right end, where $0 < k < 1$ [for example, if $w(x)$ is *uniform*, $k = \frac{1}{2}$].



(a)



(b)

Fig. 7–13

Relation between the Distributed Load and Shear. If we apply the force equation of equilibrium to the segment, then

$$+\uparrow \Sigma F_y = 0; \qquad V + w(x)\Delta x - (V + \Delta V) = 0$$
$$\Delta V = w(x)\Delta x$$

Dividing by Δx, and letting $\Delta x \to 0$, we get

$$\frac{dV}{dx} = w(x) \tag{7–1}$$

Slope of shear diagram = Distributed load intensity

If we rewrite the above equation in the form $dV = w(x)dx$ and perform an integration between any two points B and C on the beam, we see that

$$\Delta V = \int w(x)\, dx$$

$$\text{Change in shear} = \text{Area under loading curve} \tag{7–2}$$

Relation between the Shear and Moment. If we apply the moment equation of equilibrium about point O on the free-body diagram in Fig. 7–13*b*, we get

$$\zeta + \Sigma M_O = 0; \quad (M + \Delta M) - [w(x)\Delta x]\, k\Delta x - V\Delta x - M = 0$$
$$\Delta M = V\Delta x + k\, w(x)\Delta x^2$$

Dividing both sides of this equation by Δx, and letting $\Delta x \to 0$, yields

$$\frac{dM}{dx} = V$$

$$\text{Slope of moment diagram} = \text{Shear} \tag{7–3}$$

In particular, notice that a maximum bending moment $|M|_{\max}$ will occur at the point where the slope $dM/dx = 0$, since this is where the shear is equal to zero.

If Eq. 7–3 is rewritten in the form $dM = \int V\, dx$ and integrated between any two points B and C on the beam, we have

$$\Delta M = \int V\, dx$$

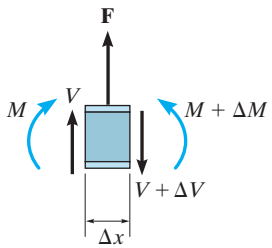
$$\text{Change in moment} = \text{Area under shear diagram} \tag{7–4}$$

As stated previously, the above equations do not apply at points where a *concentrated* force or couple moment acts. These two special cases create *discontinuities* in the shear and moment diagrams, and as a result, each deserves separate treatment.

Force. A free-body diagram of a small segment of the beam in Fig. 7–13*a*, taken from under one of the forces, is shown in Fig. 7–14*a*. Here force equilibrium requires

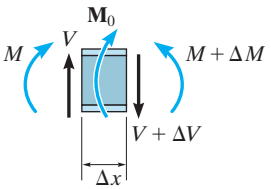
$$+\uparrow \Sigma F_y = 0; \qquad \Delta V = F \tag{7–5}$$

Since the *change in shear is positive*, the shear diagram will "jump" *upward when* **F** *acts upward* on the beam. Likewise, the jump in shear (ΔV) is downward when **F** acts downward.



(a)

Fig. 7–14



(b)

Fig. 7–14 (cont.)

Couple Moment. If we remove a segment of the beam in Fig. 7–13*a* that is located at the couple moment $\mathbf{M}_0$, the free-body diagram shown in Fig. 7–14*b* results. In this case letting $\Delta x \rightarrow 0$, moment equilibrium requires

$$\zeta + \Sigma M = 0; \qquad \Delta M = M_0 \tag{7–6}$$

Thus, the *change in moment is positive*, or the moment diagram will "jump" *upward if* $\mathbf{M}_0$ *is clockwise*. Likewise, the jump ΔM is downward when $\mathbf{M}_0$ is counterclockwise.

The examples which follow illustrate application of the above equations when used to construct the shear and moment diagrams. After working through these examples, it is recommended that you also go back and solve Examples 7.6 and 7.7 using this method.



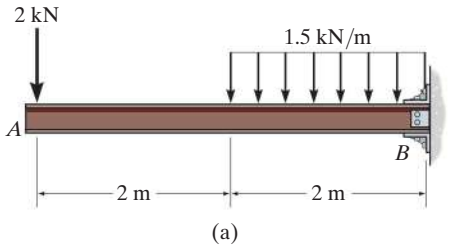
This concrete beam is used to support the deck. Its size and the placement of steel reinforcement within it can be determined once the shear and moment diagrams have been established. (© Russell C. Hibbeler)

Important Points

- The slope of the shear diagram at a point is equal to the intensity of the distributed loading, where positive distributed loading is upward, i.e., $dV/dx = w(x)$.
- The change in the shear ΔV between two points is equal to *the area* under the distributed-loading curve between the points.
- If a concentrated force acts upward on the beam, the shear will jump upward by the same amount.
- The slope of the moment diagram at a point is equal to the shear, i.e., $dM/dx = V$.
- The change in the moment ΔM between two points is equal to the *area* under the shear diagram between the two points.
- If a *clock*wise couple moment acts on the beam, the shear will not be affected; however, the moment diagram will jump *upward* by the amount of the moment.
- Points of *zero shear* represent points of *maximum or minimum moment* since $dM/dx = 0$.
- Because two integrations of $w = w(x)$ are involved to first determine the change in shear, $\Delta V = \int w(x)\,dx$, then to determine the change in moment, $\Delta M = \int V\,dx$, then if the loading curve $w = w(x)$ is a polynomial of degree n, $V = V(x)$ will be a curve of degree $n + 1$, and $M = M(x)$ will be a curve of degree $n + 2$.

EXAMPLE 7.8

Draw the shear and moment diagrams for the cantilever beam in Fig. 7–15*a*.



(a)

Fig. 7–15

SOLUTION

The support reactions at the fixed support B are shown in Fig. 7–15*b*.

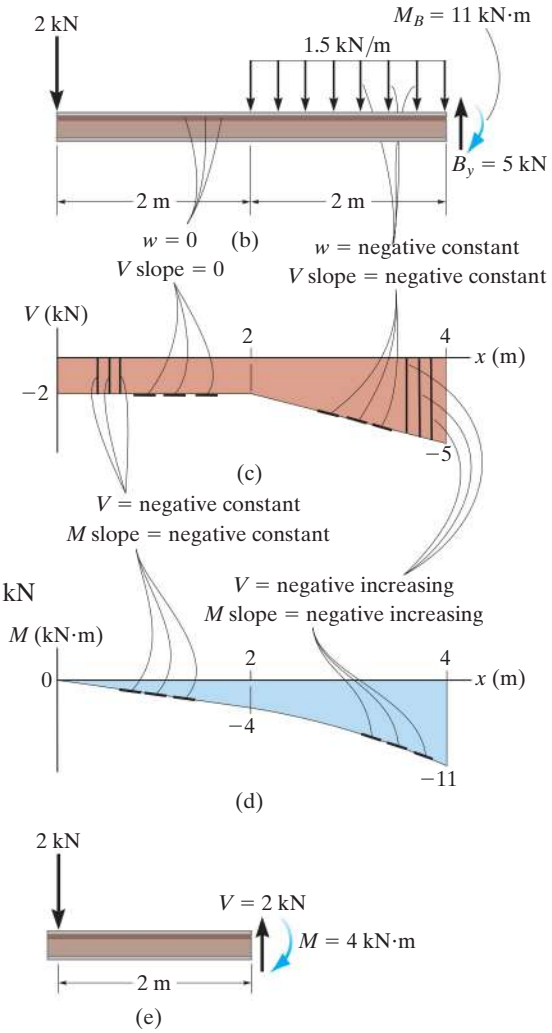
Shear Diagram. The shear at end A is -2 kN. This value is plotted at $x = 0$, Fig. 7–15*c*. Notice how the shear diagram is constructed by following the slopes defined by the loading w. The shear at $x = 4$ m is -5 kN, the reaction on the beam. This value can be verified by finding the area under the distributed loading; i.e.,

$$V|_{x=4\text{ m}} = V|_{x=2\text{ m}} + \Delta V = -2\text{ kN} - (1.5\text{ kN/m})(2\text{ m}) = -5\text{ kN}$$

Moment Diagram. The moment of zero at $x = 0$ is plotted in Fig. 7–15*d*. Construction of the moment diagram is based on knowing that its slope is equal to the shear at each point. The change of moment from $x = 0$ to $x = 2$ m is determined from the area under the shear diagram. Hence, the moment at $x = 2$ m is

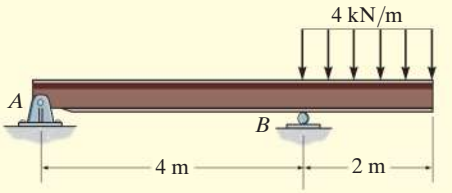
$$M|_{x=2\text{ m}} = M|_{x=0} + \Delta M = 0 + [-2\text{ kN}(2\text{ m})] = -4\text{ kN}\cdot\text{m}$$

This same value can be determined from the method of sections, Fig. 7–15*e*.



EXAMPLE 7.9

Draw the shear and moment diagrams for the overhang beam in Fig. 7–16*a*.



(a)

Fig. 7–16

SOLUTION

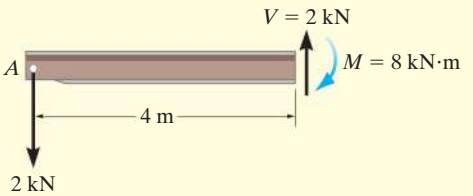
The support reactions are shown in Fig. 7–16*b*.

Shear Diagram. The shear of −2 kN at end A of the beam is plotted at $x = 0$, Fig. 7–16*c*. The slopes are determined from the loading and from this the shear diagram is constructed, as indicated in the figure. In particular, notice the positive jump of 10 kN at $x = 4$ m due to the force B_y, as indicated in the figure.

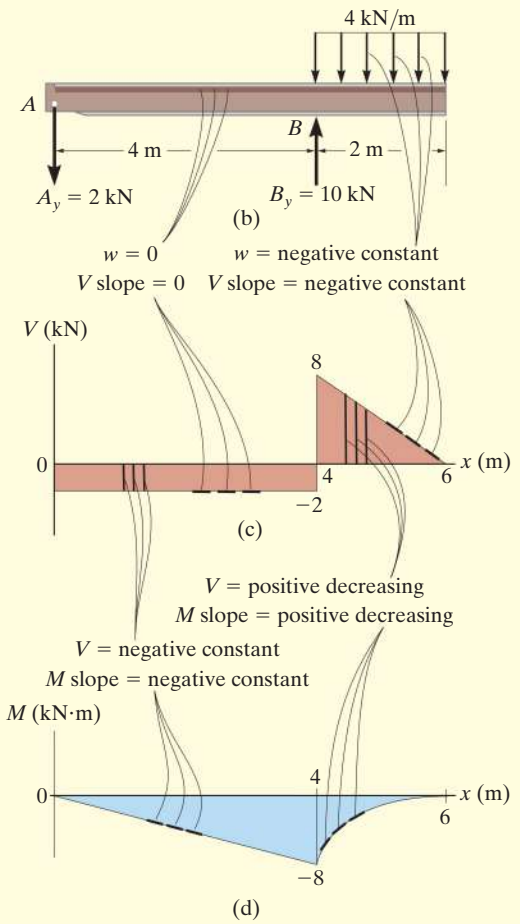
Moment Diagram. The moment of zero at $x = 0$ is plotted, Fig. 7–16*d*, then following the behavior of the slope found from the shear diagram, the moment diagram is constructed. The moment at $x = 4$ m is found from the area under the shear diagram.

$$M|_{x=4\text{ m}} = M|_{x=0} + \Delta M = 0 + [-2\text{ kN}(4\text{ m})] = -8\text{ kN}\cdot\text{m}$$

We can also obtain this value by using the method of sections, as shown in Fig. 7–16*e*.

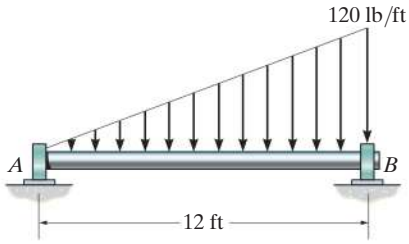


(e)



EXAMPLE 7.10

The shaft in Fig. 7–17*a* is supported by a thrust bearing at *A* and a journal bearing at *B*. Draw the shear and moment diagrams.



(a)

Fig. 7–17

SOLUTION

The support reactions are shown in Fig. 7–17*b*.

Shear Diagram. As shown in Fig. 7–17*c*, the shear at $x = 0$ is +240. Following the slope defined by the loading, the shear diagram is constructed, where at *B* its value is −480 lb. Since the shear changes sign, the point where $V = 0$ must be located. To do this we will use the method of sections. The free-body diagram of the left segment of the shaft, sectioned at an arbitrary position *x* within the region $0 \le x < 12$ ft, is shown in Fig. 7–17*e*. Notice that the intensity of the distributed load at *x* is $w = 10x$, which has been found by proportional triangles, i.e., $120/12 = w/x$.

Thus, for $V = 0$,

$$+\uparrow \Sigma F_y = 0; \qquad 240 \text{ lb} - \tfrac{1}{2}(10x)x = 0$$

$$x = 6.93 \text{ ft}$$

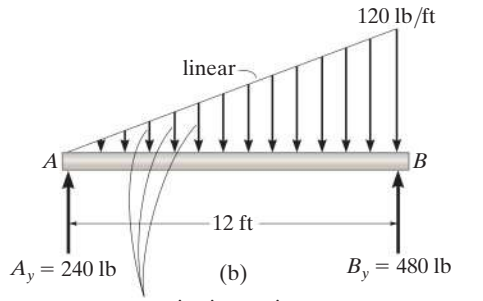
Moment Diagram. The moment diagram starts at 0 since there is no moment at *A*, then it is constructed based on the slope as determined from the shear diagram. The maximum moment occurs at $x = 6.93$ ft, where the shear is equal to zero, since $dM/dx = V = 0$, Fig. 7–17*e*,

$$\circlearrowleft + \Sigma M = 0;$$

$$M_{\max} + \tfrac{1}{2}[(10)(6.93)]\,6.93\left(\tfrac{1}{3}(6.93)\right) - 240(6.93) = 0$$

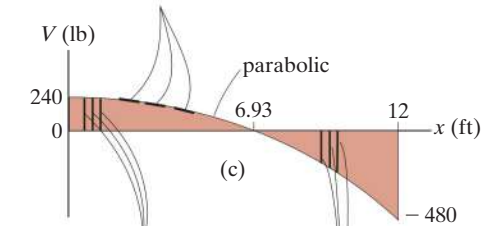
$$M_{\max} = 1109 \text{ lb} \cdot \text{ft}$$

Finally, notice how integration, first of the loading *w* which is linear, produces a shear diagram which is parabolic, and then a moment diagram which is cubic.



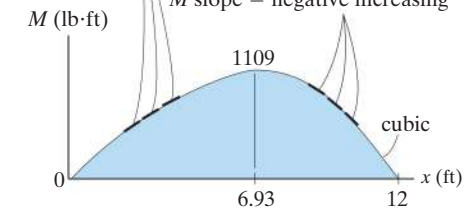
(b)

w = negative increasing
V slope = negative increasing

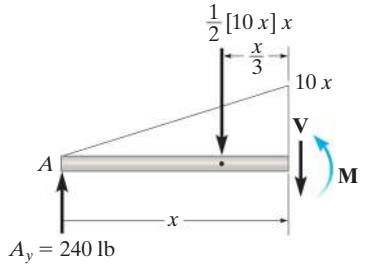


(c)

V = positive decreasing
M slope = positive decreasing
V = negative increasing
M slope = negative increasing



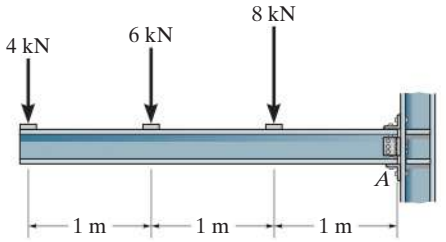
(d)



(e)

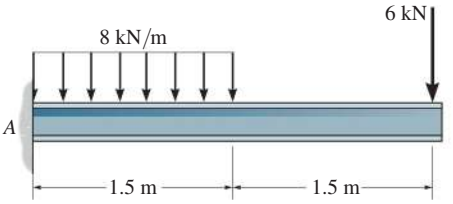
FUNDAMENTAL PROBLEMS

F7–13. Draw the shear and moment diagrams for the beam.



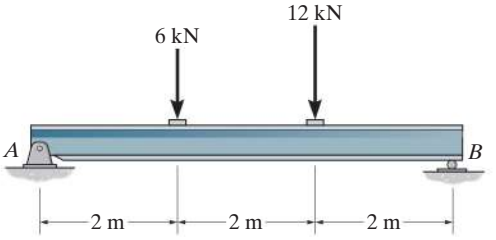
Prob. F7–13

F7–14. Draw the shear and moment diagrams for the beam.



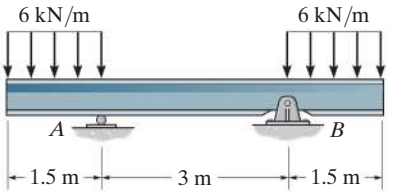
Prob. F7–14

F7–15. Draw the shear and moment diagrams for the beam.



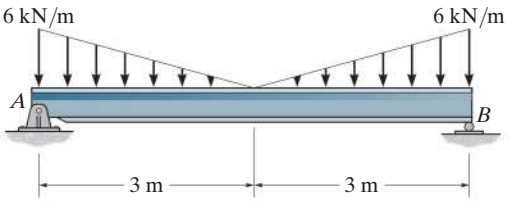
Prob. F7–15

F7–16. Draw the shear and moment diagrams for the beam.



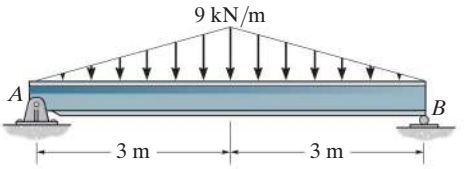
Prob. F7–16

F7–17. Draw the shear and moment diagrams for the beam.



Prob. F7–17

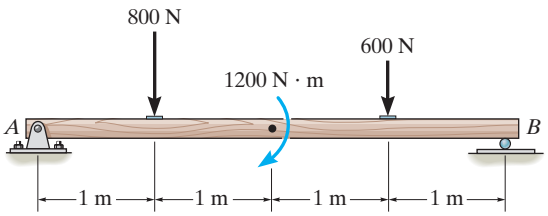
F7–18. Draw the shear and moment diagrams for the beam.



Prob. F7–18

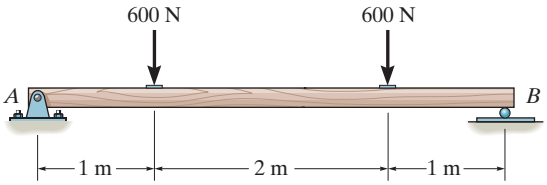
PROBLEMS

7–70. Draw the shear and moment diagrams for the beam.



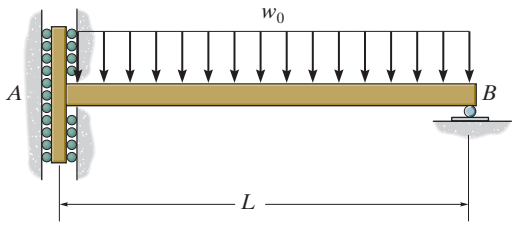
Prob. 7–70

7–71. Draw the shear and moment diagrams for the beam.



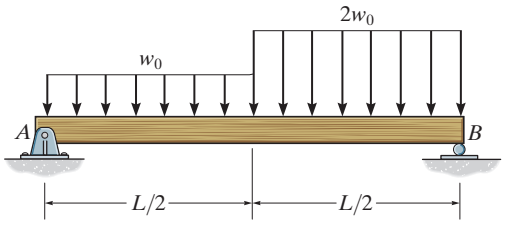
Prob. 7–71

***7–72.** Draw the shear and moment diagrams for the beam. The support at A offers no resistance to vertical load.



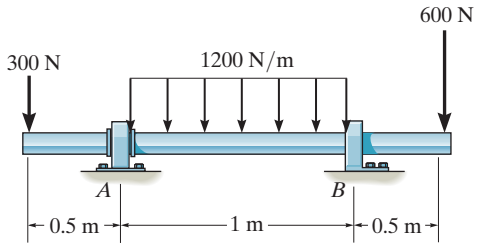
Prob. 7–72

7–73. Draw the shear and moment diagrams for the simply-supported beam.



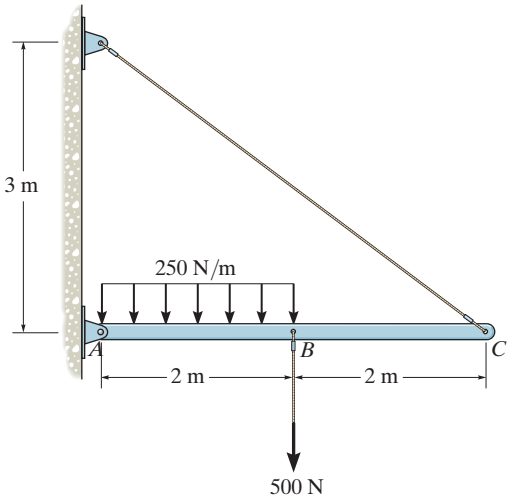
Prob. 7–73

7–74. Draw the shear and moment diagrams for the beam. The supports at A and B are a thrust bearing and journal bearing, respectively.



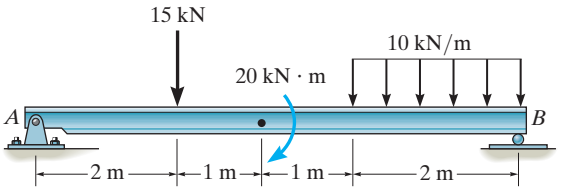
Prob. 7–74

7–75. Draw the shear and moment diagrams for the beam.



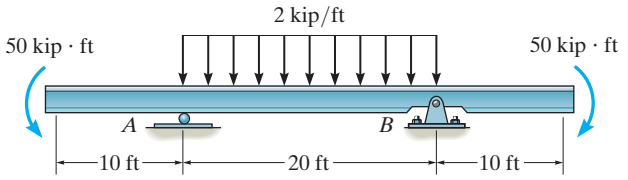
Prob. 7–75

*7–76. Draw the shear and moment diagrams for the beam.



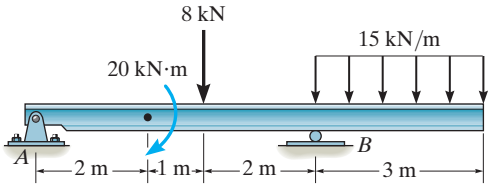
Prob. 7–76

7–77. Draw the shear and moment diagrams for the beam.



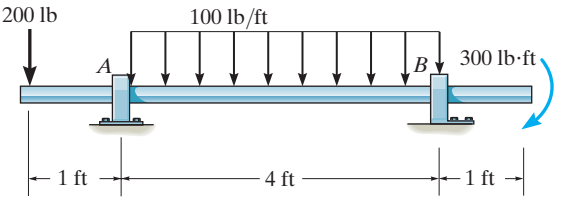
Prob. 7–77

7–78. Draw the shear and moment diagrams for the beam.



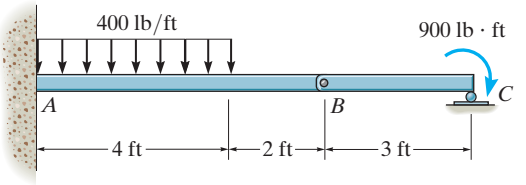
Prob. 7–78

7–79. Draw the shear and moment diagrams for the shaft. The support at *A* is a journal bearing and at *B* it is a thrust bearing.



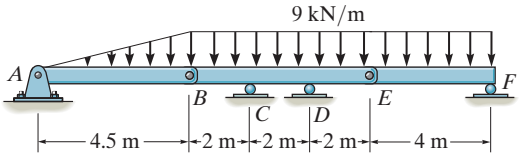
Prob. 7–79

*7–80. Draw the shear and moment diagrams for the beam.



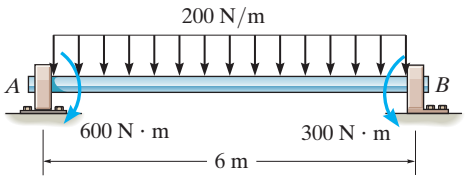
Prob. 7–80

7–81. The beam consists of three segments pin connected at *B* and *E*. Draw the shear and moment diagrams for the beam.



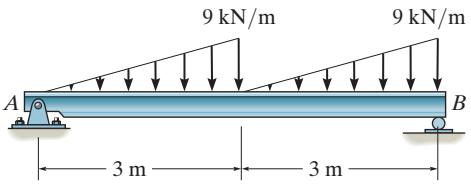
Prob. 7–81

7–82. Draw the shear and moment diagrams for the beam. The supports at A and B are a thrust and journal bearing, respectively.



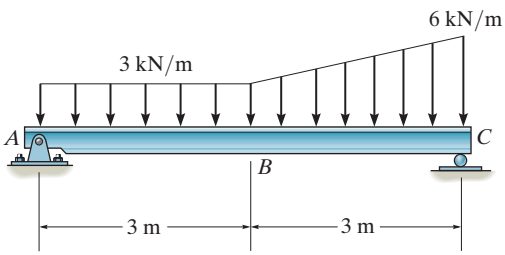
Prob. 7–82

7–83. Draw the shear and moment diagrams for the beam.



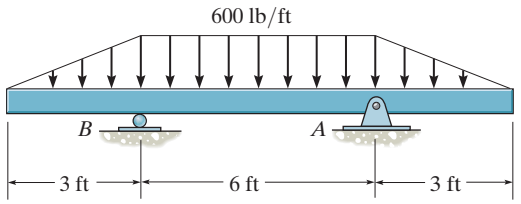
Prob. 7–83

***7–84.** Draw the shear and moment diagrams for the beam.



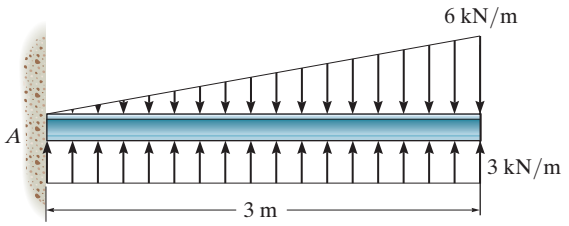
Prob. 7–84

7–85. Draw the shear and moment diagrams for the beam.



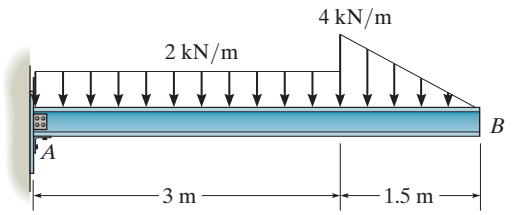
Prob. 7–85

7–86. Draw the shear and moment diagrams for the beam.



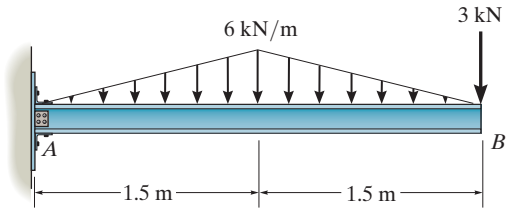
Prob. 7–86

7–87. Draw the shear and moment diagrams for the beam.



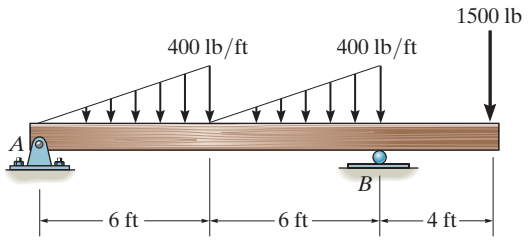
Prob. 7–87

*7–88. Draw the shear and moment diagrams for the beam.



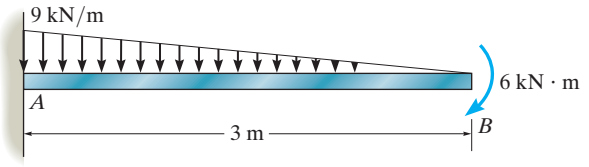
Prob. 7–88

7–89. Draw the shear and moment diagrams for the beam.



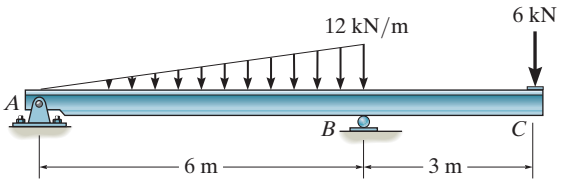
Prob. 7–89

7–90. Draw the shear and moment diagrams for the beam.



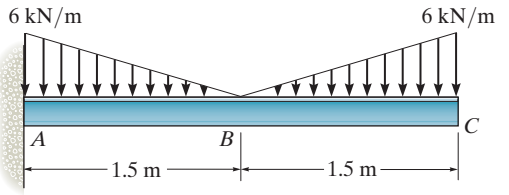
Prob. 7–90

7–91. Draw the shear and moment diagrams for the beam.



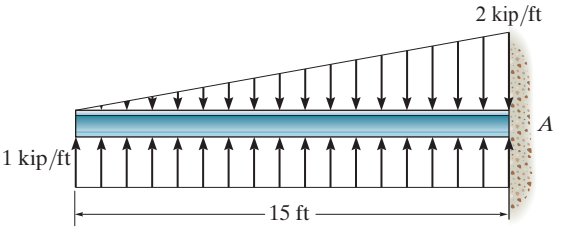
Prob. 7–91

*7–92. Draw the shear and moment diagrams for the beam.



Prob. 7–92

7–93. Draw the shear and moment diagrams for the beam.

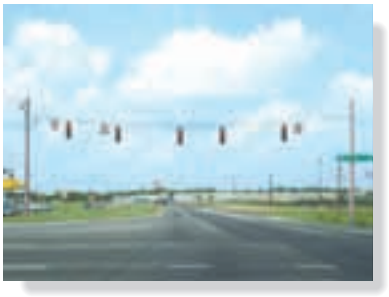


Prob. 7–93

*7.4 Cables

Flexible cables and chains combine strength with lightness and often are used in structures for support and to transmit loads from one member to another. When used to support suspension bridges and trolley wheels, cables form the main load-carrying element of the structure. In the force analysis of such systems, the weight of the cable itself may be neglected because it is often small compared to the load it carries. On the other hand, when cables are used as transmission lines and guys for radio antennas and derricks, the cable weight may become important and must be included in the structural analysis.

Three cases will be considered in the analysis that follows. In each case we will make the assumption that the cable is *perfectly flexible* and *inextensible*. Due to its flexibility, the cable offers no resistance to bending, and therefore, the tensile force acting in the cable is always tangent to the cable at points along its length. Being inextensible, the cable has a constant length both before and after the load is applied. As a result, once the load is applied, the geometry of the cable remains unchanged, and the cable or a segment of it can be treated as a rigid body.



Each of the cable segments remains approximately straight as they support the weight of these traffic lights. (© Russell C. Hibbeler)

Cable Subjected to Concentrated Loads. When a cable of negligible weight supports several concentrated loads, the cable takes the form of several straight-line segments, each of which is subjected to a constant tensile force. Consider, for example, the cable shown in Fig. 7–18, where the distances h, L_1, L_2, and L_3 and the loads $\mathbf{P}_1$ and $\mathbf{P}_2$ are known. The problem here is to determine the *nine unknowns* consisting of the tension in each of the *three* segments, the *four* components of reaction at A and B, and the *two* sags y_C and y_D at points C and D. For the solution we can write *two* equations of force equilibrium at each of points A, B, C, and D. This results in a total of *eight equations*.* To complete the solution, we need to know something about the geometry of the cable in order to obtain the necessary ninth equation. For example, if the cable's total *length* L is specified, then the Pythagorean theorem can be used to relate each of the three segmental lengths, written in terms of h, y_C, y_D, L_1, L_2, and L_3, to the total length L. Unfortunately, this type of problem cannot be solved easily by hand. Another possibility, however, is to specify one of the sags, either y_C or y_D, instead of the cable length. By doing this, the equilibrium equations are then sufficient for obtaining the unknown forces and the remaining sag. Once the sag at each point of loading is obtained, the length of the cable can then be determined by trigonometry. The following example illustrates a procedure for performing the equilibrium analysis for a problem of this type.

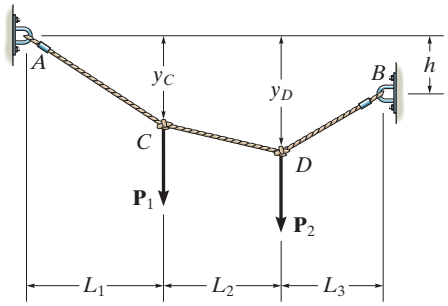
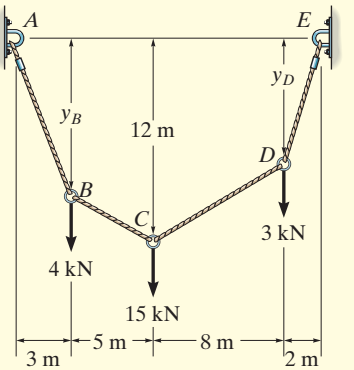


Fig. 7–18

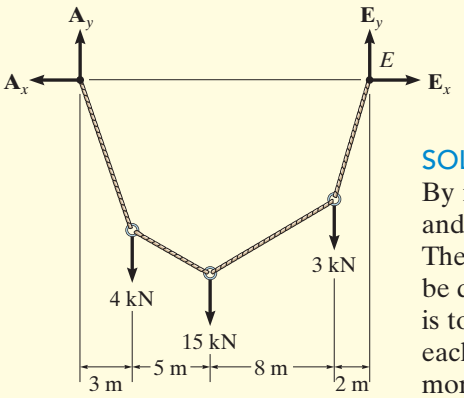
*As will be shown in the following example, the eight equilibrium equations *also* can be written for the entire cable, or any part thereof. But *no more* than *eight* independent equations are available.

EXAMPLE 7.11

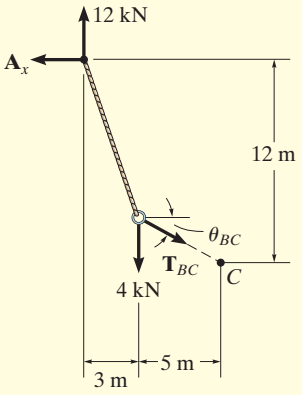
Determine the tension in each segment of the cable shown in Fig. 7–19*a*.



(a)



(b)



(c)

Fig. 7–19

SOLUTION

By inspection, there are four unknown external reactions (A_x, A_y, E_x, and E_y) and four unknown cable tensions, one in each cable segment. These eight unknowns along with the two unknown sags y_B and y_D can be determined from *ten* available equilibrium equations. One method is to apply the force equations of equilibrium ($\Sigma F_x = 0$, $\Sigma F_y = 0$) to each of the five points A through E. Here, however, we will take a more direct approach.

Consider the free-body diagram for the entire cable, Fig. 7–19*b*. Thus,

$$\pm\rightarrow \Sigma F_x = 0; \qquad -A_x + E_x = 0$$

$$\circlearrowleft +\Sigma M_E = 0;$$

$$-A_y(18\text{ m}) + 4\text{ kN}(15\text{ m}) + 15\text{ kN}(10\text{ m}) + 3\text{ kN}(2\text{ m}) = 0$$

$$A_y = 12\text{ kN}$$

$$+\uparrow \Sigma F_y = 0; \qquad 12\text{ kN} - 4\text{ kN} - 15\text{ kN} - 3\text{ kN} + E_y = 0$$

$$E_y = 10\text{ kN}$$

Since the sag $y_C = 12$ m is known, we will now consider the leftmost section, which cuts cable BC, Fig. 7–19*c*.

$$\circlearrowleft +\Sigma M_C = 0;\ A_x(12\text{ m}) - 12\text{ kN}(8\text{ m}) + 4\text{ kN}(5\text{ m}) = 0$$

$$A_x = E_x = 6.33\text{ kN}$$

$$\pm\rightarrow \Sigma F_x = 0; \qquad T_{BC}\cos\theta_{BC} - 6.33\text{ kN} = 0$$

$$+\uparrow \Sigma F_y = 0; \qquad 12\text{ kN} - 4\text{ kN} - T_{BC}\sin\theta_{BC} = 0$$

Thus,

$$\theta_{BC} = 51.6°$$

$$T_{BC} = 10.2\text{ kN} \qquad \textit{Ans.}$$

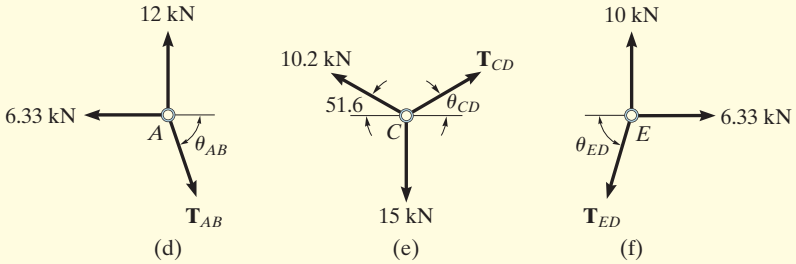


Fig. 7–19 (cont.)

Proceeding now to analyze the equilibrium of points A, C, and E in sequence, we have

Point A. (Fig. 7–19*d*).

$$\stackrel{+}{\rightarrow}\Sigma F_x = 0; \qquad T_{AB}\cos\theta_{AB} - 6.33\ \text{kN} = 0$$

$$+\uparrow\Sigma F_y = 0; \qquad -T_{AB}\sin\theta_{AB} + 12\ \text{kN} = 0$$

$$\theta_{AB} = 62.2°$$

$$T_{AB} = 13.6\ \text{kN} \qquad \textit{Ans.}$$

Point C. (Fig. 7–19*e*).

$$\stackrel{+}{\rightarrow}\Sigma F_x = 0; \qquad T_{CD}\cos\theta_{CD} - 10.2\cos 51.6°\ \text{kN} = 0$$

$$+\uparrow\Sigma F_y = 0; \qquad T_{CD}\sin\theta_{CD} + 10.2\sin 51.6°\ \text{kN} - 15\ \text{kN} = 0$$

$$\theta_{CD} = 47.9°$$

$$T_{CD} = 9.44\ \text{kN} \qquad \textit{Ans.}$$

Point E. (Fig. 7–19*f*).

$$\stackrel{+}{\rightarrow}\Sigma F_x = 0; \qquad 6.33\ \text{kN} - T_{ED}\cos\theta_{ED} = 0$$

$$+\uparrow\Sigma F_y = 0; \qquad 10\ \text{kN} - T_{ED}\sin\theta_{ED} = 0$$

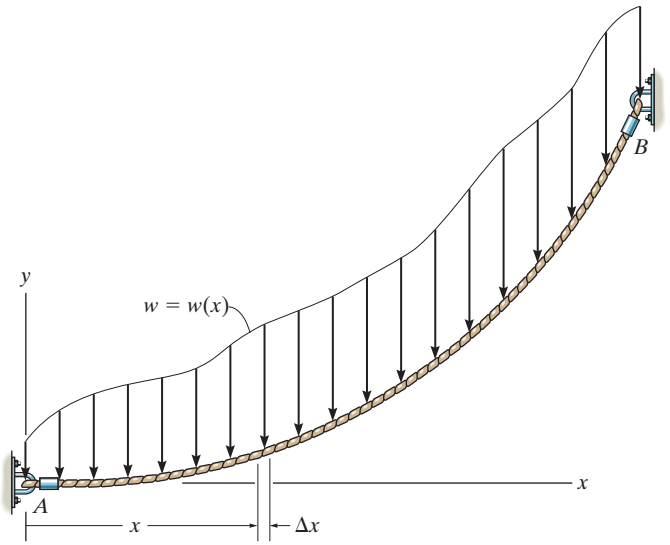
$$\theta_{ED} = 57.7°$$

$$T_{ED} = 11.8\ \text{kN} \qquad \textit{Ans.}$$

NOTE: By comparison, the maximum cable tension is in segment AB since this segment has the greatest slope (θ) and it is required that for any cable segment the horizontal component $T\cos\theta = A_x = E_x$ (a constant). Also, since the slope angles that the cable segments make with the horizontal have now been determined, it is possible to determine the sags y_B and y_D, Fig. 7–19*a*, using trigonometry.



The cable and suspenders are used to support the uniform load of a gas pipe which crosses the river. (© Russell C. Hibbeler)



(a)

Fig. 7–20

Cable Subjected to a Distributed Load. Let us now consider the weightless cable shown in Fig. 7–20*a*, which is subjected to a distributed loading $w = w(x)$ that is *measured in the x direction*. The free-body diagram of a small segment of the cable having a length Δs is shown in Fig. 7–20*b*. Since the tensile force changes in both magnitude and direction along the cable's length, we will denote this change on the free-body diagram by ΔT. Finally, the distributed load is represented by its resultant force $w(x)(\Delta x)$, which acts at a fractional distance $k(\Delta x)$ from point O, where $0 < k < 1$. Applying the equations of equilibrium, we have

$$\overset{+}{\rightarrow} \Sigma F_x = 0; \qquad -T\cos\theta + (T + \Delta T)\cos(\theta + \Delta\theta) = 0$$

$$+\uparrow \Sigma F_y = 0; \qquad -T\sin\theta - w(x)(\Delta x) + (T + \Delta T)\sin(\theta + \Delta\theta) = 0$$

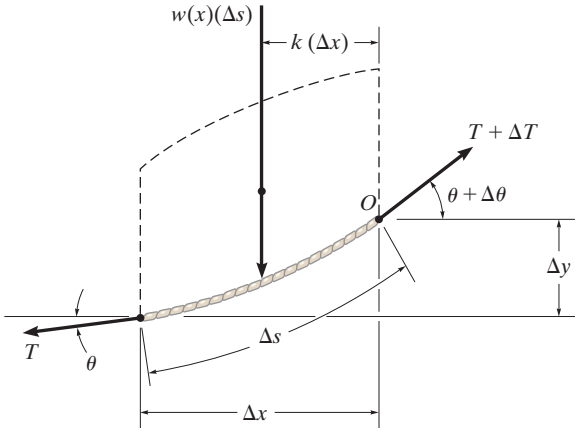
$$\circlearrowleft + \Sigma M_O = 0; \qquad w(x)(\Delta x)k(\Delta x) - T\cos\theta\,\Delta y + T\sin\theta\,\Delta x = 0$$

Dividing each of these equations by Δx and taking the limit as $\Delta x \rightarrow 0$, and therefore $\Delta y \rightarrow 0$, $\Delta\theta \rightarrow 0$, and $\Delta T \rightarrow 0$, we obtain

$$\frac{d(T\cos\theta)}{dx} = 0 \tag{7–7}$$

$$\frac{d(T\sin\theta)}{dx} - w(x) = 0 \tag{7–8}$$

$$\frac{dy}{dx} = \tan\theta \tag{7–9}$$



(b)

Fig. 7–20 (cont.)

Integrating Eq. 7–7, we have

$$T \cos \theta = \text{constant} = F_H \tag{7–10}$$

where F_H represents the horizontal component of tensile force at *any point* along the cable.

Integrating Eq. 7–8 gives

$$T \sin \theta = \int w(x)\, dx \tag{7–11}$$

Dividing Eq. 7–11 by Eq. 7–10 eliminates T. Then, using Eq. 7–9, we can obtain the slope of the cable.

$$\tan \theta = \frac{dy}{dx} = \frac{1}{F_H} \int w(x)\, dx$$

Performing a second integration yields

$$y = \frac{1}{F_H} \int \left(\int w(x)\, dx \right) dx \tag{7–12}$$

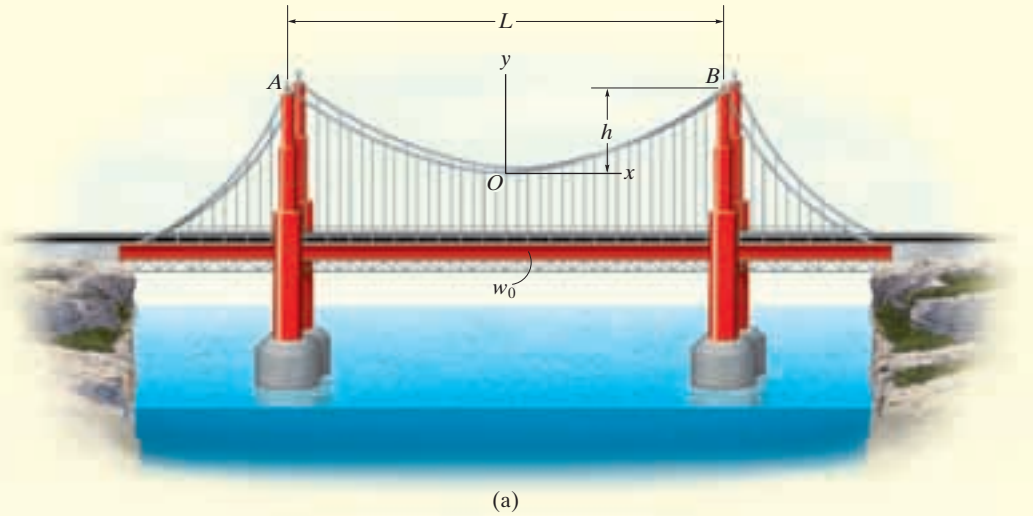
This equation is used to determine the curve for the cable, $y = f(x)$. The horizontal force component F_H and the additional two constants, say C_1 and C_2, resulting from the integration are determined by applying the boundary conditions for the curve.



The cables of the suspension bridge exert very large forces on the tower and the foundation block which have to be accounted for in their design. (© Russell C. Hibbeler)

EXAMPLE 7.12

The cable of a suspension bridge supports half of the uniform road surface between the two towers at A and B, Fig. 7–21a. If this distributed loading is w_0, determine the maximum force developed in the cable and the cable's required length. The span length L and sag h are known.



(a)

Fig. 7–21

SOLUTION

We can determine the unknowns in the problem by first finding the equation of the curve that defines the shape of the cable using Eq. 7–12. For reasons of symmetry, the origin of coordinates has been placed at the cable's center. Noting that $w(x) = w_0$, we have

$$y = \frac{1}{F_H}\int\left(\int w_0\,dx\right)dx$$

Performing the two integrations gives

$$y = \frac{1}{F_H}\left(\frac{w_0 x^2}{2} + C_1 x + C_2\right) \tag{1}$$

The constants of integration may be determined using the boundary conditions $y = 0$ at $x = 0$ and $dy/dx = 0$ at $x = 0$. Substituting into Eq. 1 and its derivative yields $C_1 = C_2 = 0$. The equation of the curve then becomes

$$y = \frac{w_0}{2F_H}x^2 \tag{2}$$

This is the equation of a *parabola*. The constant F_H may be obtained using the boundary condition $y = h$ at $x = L/2$. Thus,

$$F_H = \frac{w_0 L^2}{8h} \tag{3}$$

Therefore, Eq. 2 becomes

$$y = \frac{4h}{L^2}x^2 \tag{4}$$

Since F_H is known, the tension in the cable may now be determined using Eq. 7–10, written as $T = F_H/\cos\theta$. For $0 \le \theta < \pi/2$, the maximum tension will occur when θ is *maximum*, i.e., at point B, Fig. 7–21*a*. From Eq. 2, the slope at this point is

$$\left.\frac{dy}{dx}\right|_{x=L/2} = \tan\theta_{\max} = \left.\frac{w_0}{F_H}x\right|_{x=L/2}$$

or

$$\theta_{\max} = \tan^{-1}\left(\frac{w_0 L}{2F_H}\right) \tag{5}$$

Therefore,

$$T_{\max} = \frac{F_H}{\cos(\theta_{\max})} \tag{6}$$

Using the triangular relationship shown in Fig. 7–21*b*, which is based on Eq. 5, Eq. 6 may be written as

$$T_{\max} = \frac{\sqrt{4F_H^2 + w_0^2 L^2}}{2}$$

Substituting Eq. 3 into the above equation yields

$$T_{\max} = \frac{w_0 L}{2}\sqrt{1 + \left(\frac{L}{4h}\right)^2} \qquad \textit{Ans.}$$

For a differential segment of cable length ds, we can write

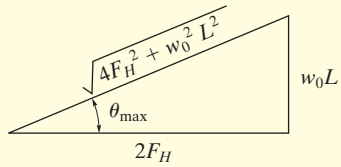
$$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$$

Hence, the total length of the cable can be determined by integration. Using Eq. 4, we have

$$\mathscr{L} = \int ds = 2\int_0^{L/2}\sqrt{1 + \left(\frac{8h}{L^2}x\right)^2}\,dx \tag{7}$$

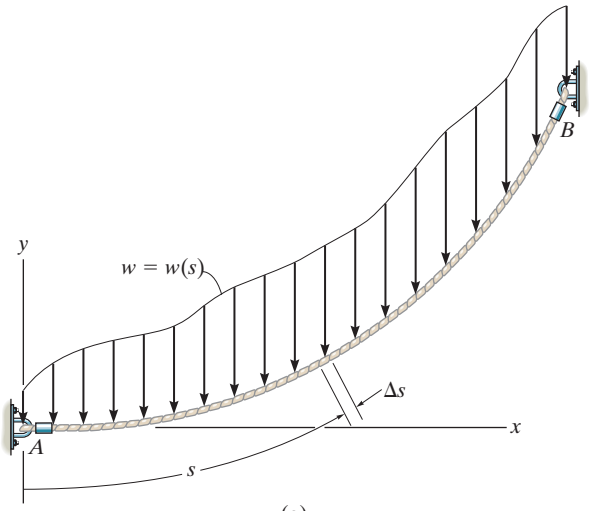
Integrating yields

$$\mathscr{L} = \frac{L}{2}\left[\sqrt{1 + \left(\frac{4h}{L}\right)^2} + \frac{L}{4h}\sinh^{-1}\left(\frac{4h}{L}\right)\right] \qquad \textit{Ans.}$$



(b)

Fig. 7–21 (cont.)



(a)

Fig. 7–22

Cable Subjected to Its Own Weight. When the weight of a cable becomes important in the force analysis, the loading function along the cable will be a function of the arc length s rather than the projected length x. To analyze this problem, we will consider a generalized loading function $w = w(s)$ acting along the cable, as shown in Fig. 7–22*a*. The free-body diagram for a small segment Δs of the cable is shown in Fig. 7–22*b*. Applying the equilibrium equations to the force system on this diagram, one obtains relationships identical to those given by Eqs. 7–7 through 7–9, but with s replacing x in Eqs. 7–7 and 7–8. Therefore, we can show that

$$T\cos\theta = F_H$$

$$T\sin\theta = \int w(s)\,ds \tag{7–13}$$

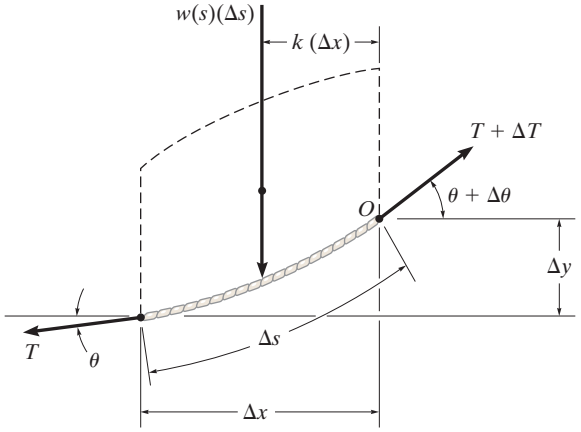
$$\frac{dy}{dx} = \frac{1}{F_H}\int w(s)\,ds \tag{7–14}$$

To perform a direct integration of Eq. 7–14, it is necessary to replace dy/dx by ds/dx. Since

$$ds = \sqrt{dx^2 + dy^2}$$

then

$$\frac{dy}{dx} = \sqrt{\left(\frac{ds}{dx}\right)^2 - 1}$$



(b)

Fig. 7–22 (cont.)

Therefore,

$$\frac{ds}{dx} = \left[1 + \frac{1}{F_H^2}\left(\int w(s)\, ds\right)^2\right]^{1/2}$$

Separating the variables and integrating we obtain

$$x = \int \frac{ds}{\left[1 + \frac{1}{F_H^2}\left(\int w(s)\, ds\right)^2\right]^{1/2}} \tag{7–15}$$

The two constants of integration, say C_1 and C_2, are found using the boundary conditions for the curve.



Electrical transmission towers must be designed to support the weights of the suspended power lines. The weight and length of the cables can be determined since they each form a catenary curve. (© Russell C. Hibbeler)

EXAMPLE 7.13

Determine the deflection curve, the length, and the maximum tension in the uniform cable shown in Fig. 7–23. The cable has a weight per unit length of $w_0 = 5\ \text{N/m}$.

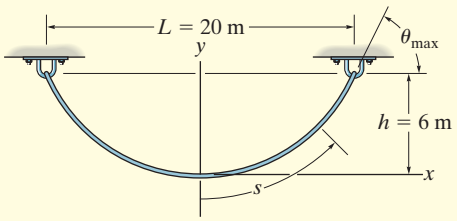


Fig. 7–23

SOLUTION

For reasons of symmetry, the origin of coordinates is located at the center of the cable. The deflection curve is expressed as $y = f(x)$. We can determine it by first applying Eq. 7–15, where $w(s) = w_0$.

$$x = \int \frac{ds}{\left[1 + (1/F_H^2)\left(\int w_0\, ds\right)^2\right]^{1/2}}$$

Integrating the term under the integral sign in the denominator, we have

$$x = \int \frac{ds}{[1 + (1/F_H^2)(w_0 s + C_1)^2]^{1/2}}$$

Substituting $u = (1/F_H)(w_0 s + C_1)$ so that $du = (w_0/F_H)\, ds$, a second integration yields

$$x = \frac{F_H}{w_0}(\sinh^{-1} u + C_2)$$

or

$$x = \frac{F_H}{w_0}\left\{\sinh^{-1}\left[\frac{1}{F_H}(w_0 s + C_1)\right] + C_2\right\} \tag{1}$$

To evaluate the constants note that, from Eq. 7–14,

$$\frac{dy}{dx} = \frac{1}{F_H}\int w_0\, ds \quad \text{or} \quad \frac{dy}{dx} = \frac{1}{F_H}(w_0 s + C_1)$$

Since $dy/dx = 0$ at $s = 0$, then $C_1 = 0$. Thus,

$$\frac{dy}{dx} = \frac{w_0 s}{F_H} \tag{2}$$

The constant C_2 may be evaluated by using the condition $s = 0$ at $x = 0$ in Eq. 1, in which case $C_2 = 0$. To obtain the deflection curve, solve for s in Eq. 1, which yields

$$s = \frac{F_H}{w_0}\sinh\left(\frac{w_0}{F_H}x\right) \tag{3}$$

Now substitute into Eq. 2, in which case

$$\frac{dy}{dx} = \sinh\left(\frac{w_0}{F_H}x\right)$$

Hence,

$$y = \frac{F_H}{w_0}\cosh\left(\frac{w_0}{F_H}x\right) + C_3$$

If the boundary condition $y = 0$ at $x = 0$ is applied, the constant $C_3 = -F_H/w_0$, and therefore the deflection curve becomes

$$y = \frac{F_H}{w_0}\left[\cosh\left(\frac{w_0}{F_H}x\right) - 1\right] \tag{4}$$

This equation defines the shape of a ***catenary curve***. The constant F_H is obtained by using the boundary condition that $y = h$ at $x = L/2$, in which case

$$h = \frac{F_H}{w_0}\left[\cosh\left(\frac{w_0 L}{2F_H}\right) - 1\right] \tag{5}$$

Since $w_0 = 5$ N/m, $h = 6$ m, and $L = 20$ m, Eqs. 4 and 5 become

$$y = \frac{F_H}{5\text{ N/m}}\left[\cosh\left(\frac{5\text{ N/m}}{F_H}x\right) - 1\right] \tag{6}$$

$$6\text{ m} = \frac{F_H}{5\text{ N/m}}\left[\cosh\left(\frac{50\text{ N}}{F_H}\right) - 1\right] \tag{7}$$

Equation 7 can be solved for F_H by using a trial-and-error procedure. The result is

$$F_H = 45.9\text{ N}$$

and therefore the deflection curve, Eq. 6, becomes

$$y = 9.19[\cosh(0.109x) - 1]\text{ m} \qquad \textit{Ans.}$$

Using Eq. 3, with $x = 10$ m, the half-length of the cable is

$$\frac{\mathscr{L}}{2} = \frac{45.9\text{ N}}{5\text{ N/m}}\sinh\left[\frac{5\text{ N/m}}{45.9\text{ N}}(10\text{ m})\right] = 12.1\text{ m}$$

Hence,

$$\mathscr{L} = 24.2\text{ m} \qquad \textit{Ans.}$$

Since $T = F_H/\cos\theta$, the maximum tension occurs when θ is maximum, i.e., at $s = \mathscr{L}/2 = 12.1$ m. Using Eq. 2 yields

$$\left.\frac{dy}{dx}\right|_{s=12.1\text{ m}} = \tan\theta_{\max} = \frac{5\text{ N/m}(12.1\text{ m})}{45.9\text{ N}} = 1.32$$

$$\theta_{\max} = 52.8°$$

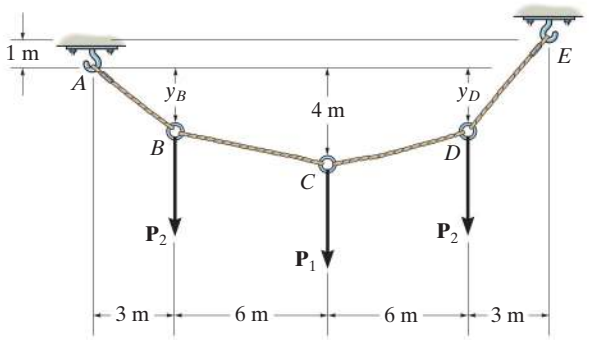
And so,

$$T_{\max} = \frac{F_H}{\cos\theta_{\max}} = \frac{45.9\text{ N}}{\cos 52.8°} = 75.9\text{ N} \qquad \textit{Ans.}$$

PROBLEMS

7–94. The cable supports the three loads shown. Determine the sags y_B and y_D of B and D. Take $P_1 = 800$ N, $P_2 = 500$ N.

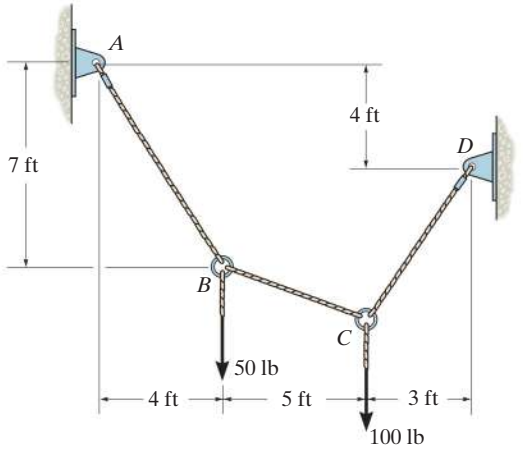
7–95. The cable supports the three loads shown. Determine the magnitude of $\mathbf{P}_1$ if $P_2 = 600$ N and $y_B = 3$ m. Also find sag y_D.



Probs. 7–94/95

7

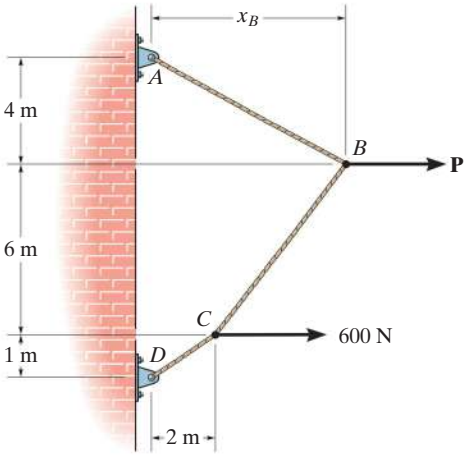
***7–96.** Determine the tension in each segment of the cable and the cable's total length.



Prob. 7–96

7–97. The cable supports the loading shown. Determine the distance x_B the force at B acts from A. Set $P = 800$ N.

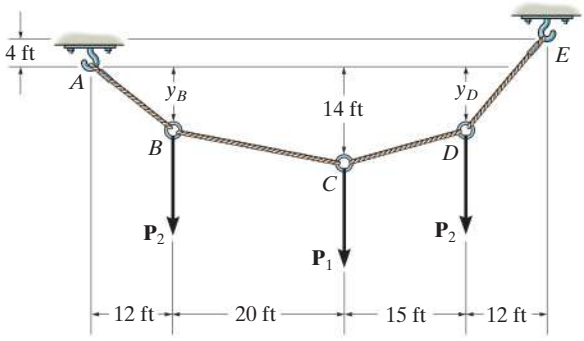
7–98. The cable supports the loading shown. Determine the magnitude of the horizontal force $\mathbf{P}$ so that $x_B = 5$ m.



Probs. 7–97/98

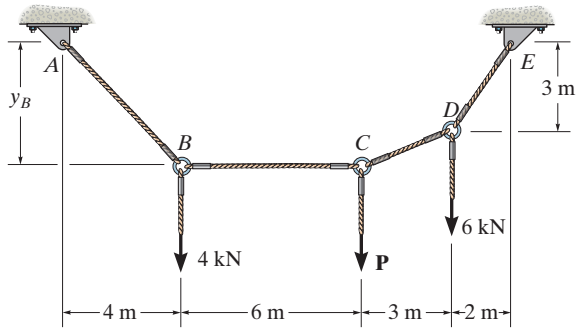
7–99. The cable supports the three loads shown. Determine the sags y_B and y_D of points B and D. Take $P_1 = 400$ lb, $P_2 = 250$ lb.

***7–100.** The cable supports the three loads shown. Determine the magnitude of $\mathbf{P}_1$ if $P_2 = 300$ lb and $y_B = 8$ ft. Also find the sag y_D.



Probs. 7–99/100

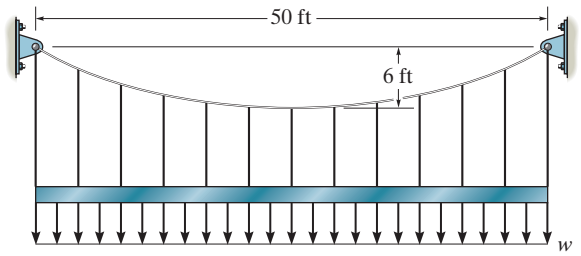
7–101. Determine the force P needed to hold the cable in the position shown, i.e., so segment BC remains horizontal. Also, compute the sag y_B and the maximum tension in the cable.



Prob. 7–101

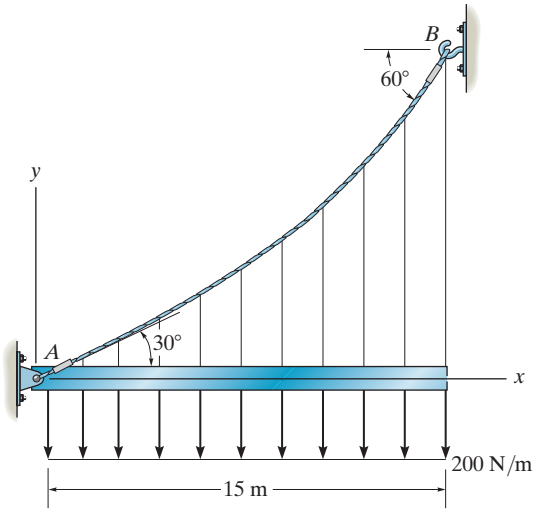
7–102. Determine the maximum uniform loading w, measured in lb/ft, that the cable can support if it is capable of sustaining a maximum tension of 3000 lb before it will break.

7–103. The cable is subjected to a uniform loading of $w = 250$ lb/ft. Determine the maximum and minimum tension in the cable.



Probs. 7–102/103

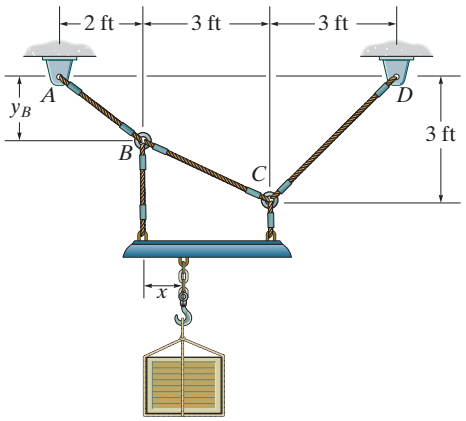
***7–104.** The cable AB is subjected to a uniform loading of 200 N/m. If the weight of the cable is neglected and the slope angles at points A and B are 30° and 60°, respectively, determine the curve that defines the cable shape and the maximum tension developed in the cable.



Prob. 7–104

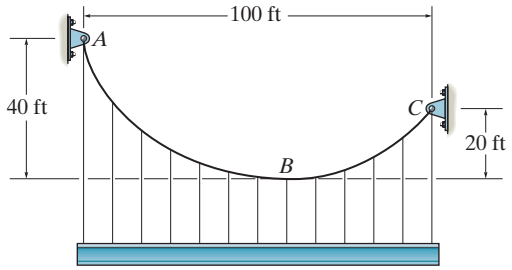
7–105. If $x = 2$ ft and the crate weighs 300 lb, which cable segment AB, BC, or CD has the greatest tension? What is this force and what is the sag y_B?

7–106. If $y_B = 1.5$ ft, determine the largest weight of the crate and its placement x so that neither cable segment AB, BC, or CD is subjected to a tension that exceeds 200 lb.



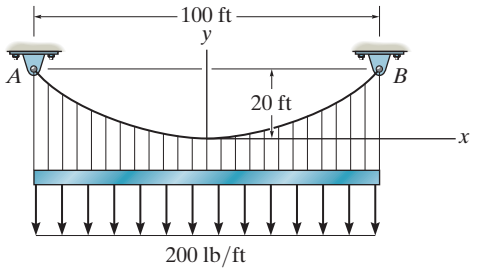
Probs. 7–105/106

7–107. The cable supports a girder which weighs 850 lb/ft. Determine the tension in the cable at points A, B, and C.



Prob. 7–107

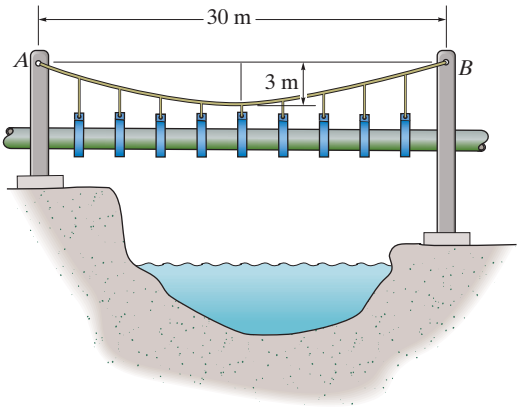
***7–108.** The cable is subjected to a uniform loading of $w = 200$ lb/ft. Determine the maximum and minimum tension in the cable.



Prob. 7–108

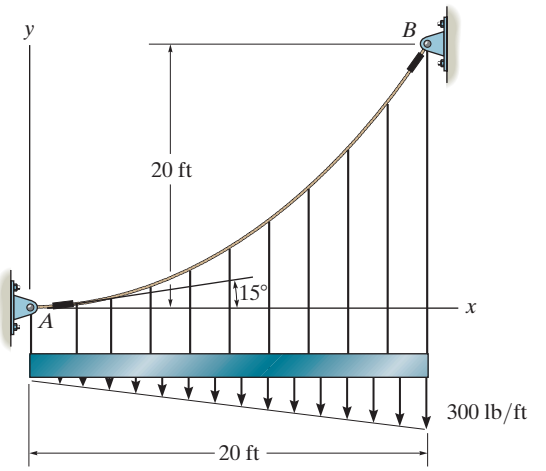
7–109. If the pipe has a mass per unit length of 1500 kg/m, determine the maximum tension developed in the cable.

7–110. If the pipe has a mass per unit length of 1500 kg/m, determine the minimum tension developed in the cable.



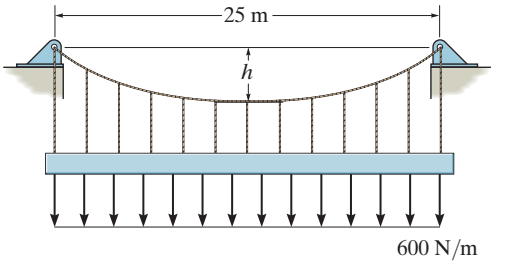
Probs. 7–109/110

7–111. Determine the maximum tension developed in the cable if it is subjected to the triangular distributed load.



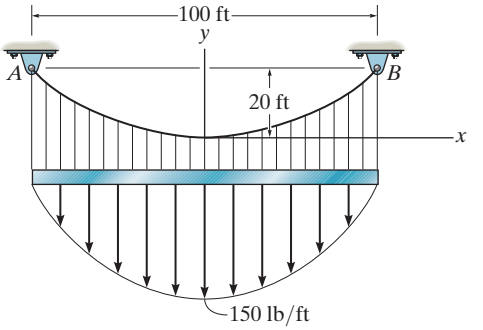
Prob. 7–111

***7–112.** The cable will break when the maximum tension reaches $T_{max} = 10$ kN. Determine the minimum sag h if it supports the uniform distributed load of $w = 600$ N/m.



Prob. 7–112

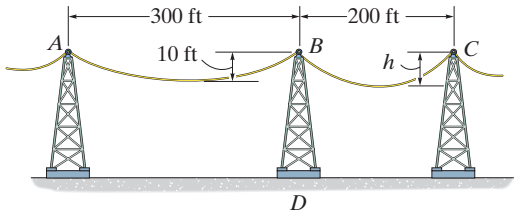
7–113. The cable is subjected to the parabolic loading $w = 150(1 - (x/50)^2)$ lb/ft, where x is in ft. Determine the equation $y = f(x)$ which defines the cable shape AB and the maximum tension in the cable.



Prob. 7–113

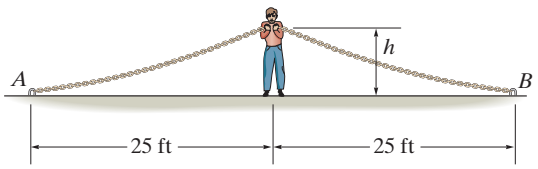
7–114. The power transmission cable weighs 10 lb/ft. If the resultant horizontal force on tower *BD* is required to be zero, determine the sag *h* of cable *BC*.

7–115. The power transmission cable weighs 10 lb/ft. If $h = 10$ ft, determine the resultant horizontal and vertical forces the cables exert on tower *BD*.



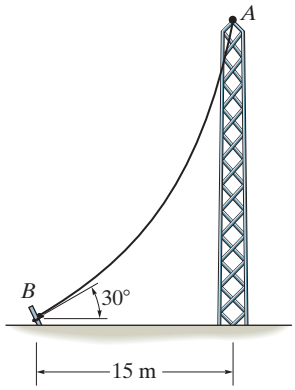
Probs. 7–114/115

***7–116.** The man picks up the 52-ft chain and holds it just high enough so it is completely off the ground. The chain has points of attachment *A* and *B* that are 50 ft apart. If the chain has a weight of 3 lb/ft, and the man weighs 150 lb, determine the force he exerts on the ground. Also, how high *h* must he lift the chain? *Hint*: The slopes at *A* and *B* are zero.



Prob. 7–116

7–117. The cable has a mass of 0.5 kg/m and is 25 m long. Determine the vertical and horizontal components of force it exerts on the top of the tower.



Prob. 7–117

7–118. A 50-ft cable is suspended between two points a distance of 15 ft apart and at the same elevation. If the minimum tension in the cable is 200 lb, determine the total weight of the cable and the maximum tension developed in the cable.

7–119. Show that the deflection curve of the cable discussed in Example 7.13 reduces to Eq. 4 in Example 7.12 when the *hyperbolic cosine function* is expanded in terms of a series and only the first two terms are retained. (The answer indicates that the *catenary* may be replaced by a *parabola* in the analysis of problems in which the sag is small. In this case, the cable weight is assumed to be uniformly distributed along the horizontal.)

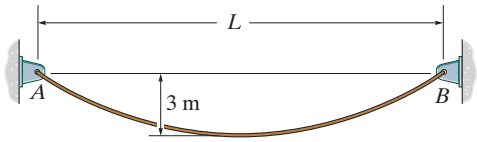
***7–120.** A telephone line (cable) stretches between two points which are 150 ft apart and at the same elevation. The line sags 5 ft and the cable has a weight of 0.3 lb/ft. Determine the length of the cable and the maximum tension in the cable.

7–121. A cable has a weight of 2 lb/ft. If it can span 100 ft and has a sag of 12 ft, determine the length of the cable. The ends of the cable are supported from the same elevation.

7–122. A cable has a weight of 3 lb/ft and is supported at points that are 500 ft apart and at the same elevation. If it has a length of 600 ft, determine the sag.

7–123. A cable has a weight of 5 lb/ft. If it can span 300 ft and has a sag of 15 ft, determine the length of the cable. The ends of the cable are supported at the same elevation.

***7–124.** The 10 kg/m cable is suspended between the supports *A* and *B*. If the cable can sustain a maximum tension of 1.5 kN and the maximum sag is 3 m, determine the maximum distance *L* between the supports.

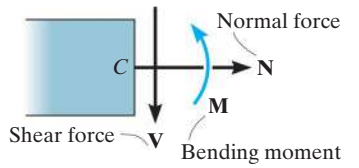


Prob. 7–124

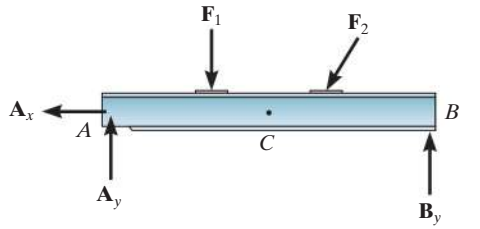
CHAPTER REVIEW

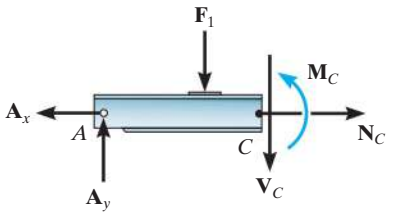
Internal Loadings

If a coplanar force system acts on a member, then in general a resultant internal *normal force* **N**, *shear force* **V**, and *bending moment* **M** will act at any cross section along the member. For two-dimensional problems the positive directions of these loadings are shown in the figure.



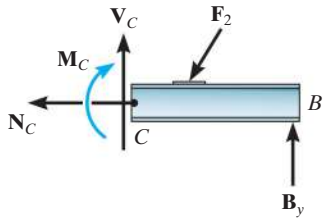
The resultant internal normal force, shear force, and bending moment are determined using the method of sections. To find them, the member is sectioned at the point C where the internal loadings are to be determined. A free-body diagram of one of the sectioned parts is then drawn and the internal loadings are shown in their positive directions.



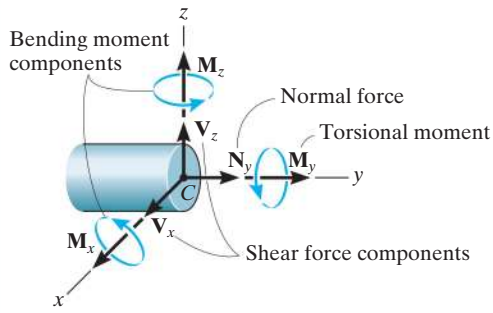


The resultant normal force is determined by summing forces normal to the cross section. The resultant shear force is found by summing forces tangent to the cross section, and the resultant bending moment is found by summing moments about the geometric center or centroid of the cross-sectional area.

$$\Sigma F_x = 0$$
$$\Sigma F_y = 0$$
$$\Sigma M_C = 0$$

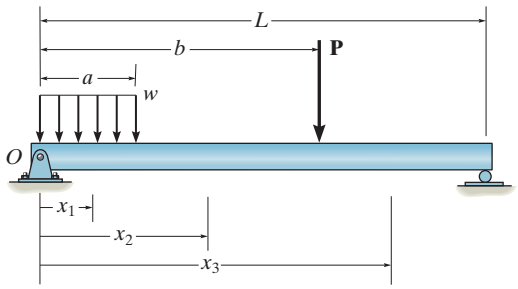


If the member is subjected to a three-dimensional loading, then, in general, a *torsional moment* will also act on the cross section. It can be determined by summing moments about an axis that is perpendicular to the cross section and passes through its centroid.



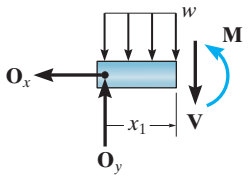
Shear and Moment Diagrams

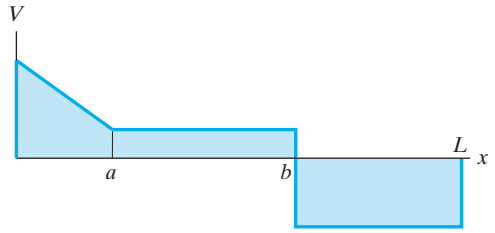
To construct the shear and moment diagrams for a member, it is necessary to section the member at an arbitrary point, located a distance x from the left end.



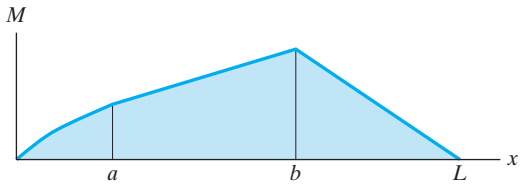
If the external loading consists of changes in the distributed load, or a series of concentrated forces and couple moments act on the member, then different expressions for V and M must be determined within regions between any load discontinuities.

The unknown shear and moment are indicated on the cross section in the positive direction according to the established sign convention, and then the internal shear and moment are determined as functions of x.





Each of the functions of the shear and moment is then plotted to create the shear and moment diagrams.



Relations between Shear and Moment

It is possible to plot the shear and moment diagrams quickly by using differential relationships that exist between the distributed loading *w*, *V* and *M*.

The slope of the shear diagram is equal to the distributed loading at any point. The slope is positive if the distributed load acts upward, and vice-versa.

The slope of the moment diagram is equal to the shear at any point. The slope is positive if the shear is positive, or vice-versa.

The change in shear between any two points is equal to the area under the distributed loading between the points.

The change in the moment is equal to the area under the shear diagram between the points.

$$\frac{dV}{dx} = w$$

$$\frac{dM}{dx} = V$$

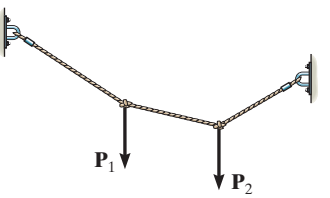
$$\Delta V = \int w\,dx$$

$$\Delta M = \int V\,dx$$

Cables

When a flexible and inextensible cable is subjected to a series of concentrated forces, then the analysis of the cable can be performed by using the equations of equilibrium applied to free-body diagrams of either segments or points of application of the loading.

If external distributed loads or the weight of the cable are to be considered, then the shape of the cable must be determined by first analyzing the forces on a differential segment of the cable and then integrating this result. The two constants, say C_1 and C_2, resulting from the integration are determined by applying the boundary conditions for the cable.



$$y = \frac{1}{F_H}\int\left(\int w(x)\,dx\right)dx$$

Distributed load

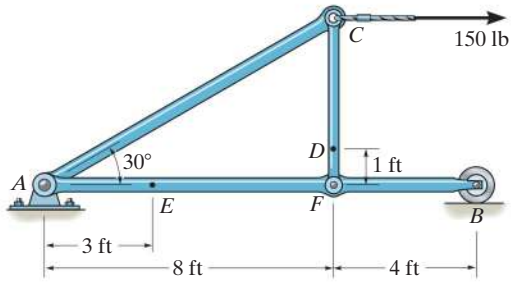
$$x = \int \frac{ds}{\left[1 + \frac{1}{F_H^2}\left(\int w(s)\,ds\right)^2\right]^{1/2}}$$

Cable weight

REVIEW PROBLEMS

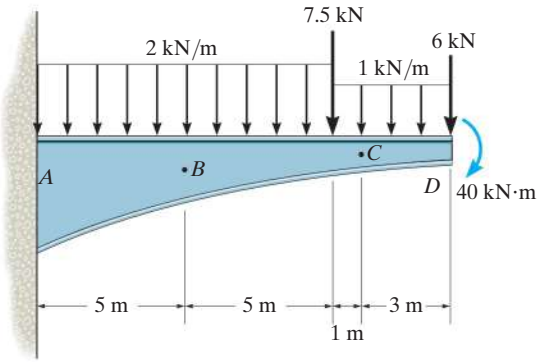
All problem solutions must include FBDs.

R7–1. Determine the internal normal force, shear force, and moment at points *D* and *E* of the frame.



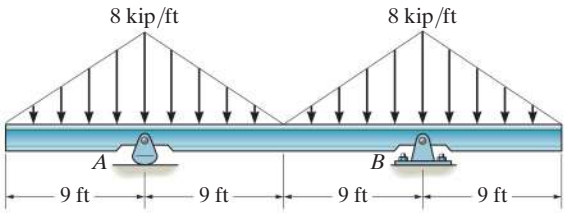
Prob. R7–1

R7–2. Determine the normal force, shear force, and moment at points *B* and *C* of the beam.



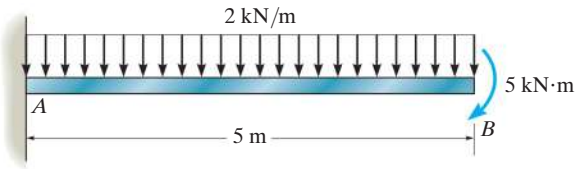
Prob. R7–2

R7–3. Draw the shear and moment diagrams for the beam.



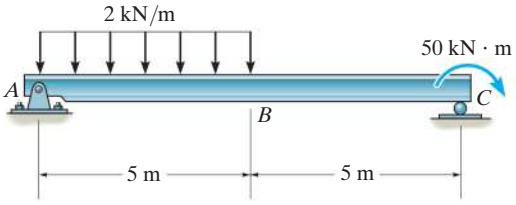
Prob. R7–3

R7–4. Draw the shear and moment diagrams for the beam.



Prob. R7–4

R7–5. Draw the shear and moment diagrams for the beam.



Prob. R7–5

R7–6. A chain is suspended between points at the same elevation and spaced a distance of 60 ft apart. If it has a weight per unit length of 0.5 lb/ft and the sag is 3 ft, determine the maximum tension in the chain.